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Is self-management a burden? What are the experiences of women self-managing chronic conditions during pregnancy: a systematic review

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Abstract

Objective

This systematic review examines the qualitative literature on women's experiences of self-managing chronic conditions in pregnancy.

Design

Systematic review of qualitative literature. Searches were performed in PubMed and CINAHL from inception to February 2021. Critical interpretive synthesis informed the coding framework and the analysis of the data. The Burden of Treatment theory emerged during the initial analysis as having the most synergy with the included literature, themes were refined to consider key concepts from this theory.

Participants

Pregnant women who are self-managing a chronic condition.

Results

A total of 2,695 articles were screened and 25 were reviewed in detail. All 16 included studies concerned diabetes self-management in pregnancy. Common themes coalesced around motivations for, and barriers to, self-management. Women self-managed primarily for the health of their baby. Barriers identified were anxiety, lack of understanding, and a lack of support from families and healthcare professionals.

Conclusions

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Pregnant women have different motivating factors for self-management than the general population and further research on a range of self-management of chronic conditions in pregnancy is needed.

Prospero Registration Number

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Keywords: Systematic reviews, pregnancy, diabetes in pregnancy, chronic conditions, qualitative research, antenatal care, self-management, self-care, self-efficacy

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Article Summary

Short Title: Is self-management a burden?

Strengths and limitations of this study

- Pregnant women's experiences of self-managing during pregnancy is under researched compared to the general population. Lived experiences are often over looked in self-management literature and by using qualitative literature we have been able to address some of these research gaps.
- Limitations include that the coding framework is constrained by how the original researchers interpreted the interviews and their interactions with patients.
- Distinguishing between pre-existing chronic conditions and pregnancy acquired chronic conditions is important, but it was not a distinction that was reflected in the dataset; it is both a limitation of the included studies and this review.
- The burden of treatment theory does not perfectly map to pregnancy, however elements of this theory are helpful in understanding how and why patients interact with self-management.

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Introduction

Chronic diseases that manifest during pregnancy, or exist pre-pregnancy, can have a lasting impact on the health of the mother (1). Numerous conditions can affect women during pregnancy, some confined to that time (for example pre-eclampsia or gestational diabetes) and others that are longer lasting (chronic hypertension and diabetes). These conditions may lead to harm for both the mother and the baby; hypertensive disorders, for example, are one of the leading causes of maternal mortality globally (2).

Self-management

Self-management has been shown to support and improve the management of chronic conditions in the general population and there is a growing body of research to suggest self-management of chronic conditions in the pregnant population is also effective (3) (4) (5). Self-management interventions in pregnancy tend to focus on self-monitoring (blood pressure, blood glucose etc.) and in some cases titrating medication (insulin). Self-management following a hypertensive pregnancy has been explored post-partum, the evidence to date suggests not only is it feasible, but it may result in better diastolic blood pressure control at six months post-partum (4). In the antenatal period, the OPTIMUM-BP pilot trial suggested that self-monitoring of blood pressure was both feasible and acceptable to women with chronic hypertension (5). Self-management of diabetes in pregnancy has also been explored. In pregnant women with type 1 and type 2 diabetes, self-monitoring of blood glucose is common (6) and self-management programmes in women with gestational diabetes are also becoming more commonplace (3) (7). In the general population, self-management has been explored in numerous chronic conditions; in particular hypertension and diabetes, where there is a large body of research around understanding the patient experience (8) (9) (10). Much of this research has demonstrated patient's confidence and ability to self-manage

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(sometimes referred to as self-efficacy) (8) and suggests that self-management strategies improve health outcomes for those with chronic conditions (11). These conditions also affect women during pregnancy, but thus far this population are under-represented in this field of research. Moreover, research on self-management in the general population may not be wholly applicable to the pregnant population. Pregnancy presents a different set of motivations for self-management, not accounted for in current self-management frameworks which often assume long-term self-management, whereas with pregnancy the period of management is temporarily bounded even if it continues post-partum and considerations are for the mother and baby dyad.

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Methods

We took a critical interpretative synthesis (CIS) approach (12). This allowed us to iteratively develop the research question and the inclusion criteria as the search progressed and the lack of research on the self-management of chronic conditions in pregnancy became apparent. In light of a lack of data, it was necessary to limit the scope of the review to the following research question: "what are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?"

We registered this review in PROSPERO (CRD42019136681, see online Appendix 1). Searches were performed in PubMed and CINAHL [EBSCOHost] (1982-present) from inception to February 2021 (see Appendix 1).

Potentially eligible studies were screened by two independent reviewers against the following inclusion criteria:

1. Chronic condition: including but not exclusively diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease.
2. Pregnancy.
3. Self-management: including but not exclusively self-monitoring, self-weighing, self-care.
4. Qualitative methods.

This review used a broad definition of self-management, "the ability of an individual, in conjunction with family, community and health care professionals, to manage symptoms, treatments and lifestyle changes" (13). We expanded the search criteria retrospectively to include risk factors for chronic conditions, in order to capture as much data as possible. The authors felt that qualitative literature was best placed to explore the lived experiences of people self-managing during

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pregnancy. Papers were excluded if the reviewers concluded that commonly recognised qualitative methods were not used, i.e. interviews, focus groups, or ethnographic observations.

Two reviewers (BJ & JK) independently reviewed the titles and abstracts of identified articles, a full text assessment of the relevant papers by both reviewers followed. The reference lists of included studies were screened but found no further articles to include. Disagreements on articles were resolved by consensus or discussion with a third reviewer (KT). Data extraction was conducted by BJ and checked by the study team. It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Data Analysis

CIS also informed the analysis of the data (12). Analysis began with a detailed examination of the included papers, identifying themes and developing an understanding of the material. Following the CIS approach, themes were refined in relation to Burden of Treatment theory which emerged during the initial analysis as having the most synergy with the included literature. The coding framework was revised considering this theory's key concepts, such as the effect of patient work on self-management efficacy and the role of support networks, allowing for a more nuanced interpretation of the woman's experience of self-management. As one of the fundamental principles of a CIS approach is to question 'taken-for-granted assumptions' (12) the coding framework was also developed to critique the idea that general population self-management research was applicable to pregnancy. Data extraction was conducted by BJ using NVivo 12, and a coding framework was developed by BJ and agreed upon by the study team (LH, KT, RM).

The Burden of Treatment

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May et al have developed a theory explaining the relationship between sick people and their healthcare providers in the context of chronic disease. In the absence of a possible cure, patients must engage in 'routine work' to manage the disease. Self-management is the 'work' patients and those in their support networks take up. May et al describe patient work as "symptom monitoring, treatment management, and lifestyle changes" while "maintaining [these tasks] alongside the demands of other aspects of everyday life" (14). Transferring the burden of treatment to the patient prompts a change in the patient-clinician relationship. Within pregnancy this could theoretically provide women with more agency, as well as more responsibility.

The first interpretation of the data, reviewing the themes that already existed in the included papers, revealed a synergy with Burden of Treatment theory; therefore, the coding framework was iteratively mapped to two broad themes reflecting elements of the burden of treatment. These themes were grouped around motivations for self-management and barriers to self-management. As the coding progressed, more elements of the theory were incorporated into the analysis. May claims "people and members of their social networks are expected to identify, understand and explain the diverse tasks that make up their work" (14), and goes on to call this process 'sense making'. This idea was reflected in the themes concerning support from health care professionals and family, the impact of a lack of support, and women's anxiety and understanding of their diagnosis.

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Results

Searches in PubMed and CINAHL identified 2,745 articles, which when controlled for duplicates left 2,695 for screening (see Appendix 1 and Figure 1 for further details). 25 articles were identified by the two reviewers for full text review, of which 16 fulfilled the inclusion criteria (see Table 1). Data quality checks using the COREQ guidelines were carried out (see online Appendix 2). Fifteen of the included studies were of medium to high quality. One was considered low quality because of the small sample size.

Included studies were published between 2011 and 2020 and all focused on diabetes in pregnancy. Six were conducted in the United States, four in Australia, and the remaining studies were carried out in Canada, Spain, Thailand, China, Singapore and New Zealand. The studies recruited from a range of settings: gestational diabetes clinics, antenatal clinics supporting low-income women, health care provider's offices, day care centres, and hospital based obstetric clinics. Fifteen of the studies used one-to-one interviews and one used focus groups for primary data collection. Two studies supplemented the interviews with focus groups, one supplemented with additional diary entries. One study was a mixed methods study, including one-to-one interviews alongside survey data. The study population for all included studies was pregnant women, although one study had a mixed population of women who were currently pregnant, women planning on getting pregnant, and women who had previously been pregnant. Only papers concerning diabetes self-management in pregnancy fulfilled the inclusion criteria, leading to the results focusing on the challenges of self-management in this disease.

The two broad themes that emerged from the data were motivations for and, conversely, barriers to self-management (see Table 2). Under the former, women express that having a healthy baby,

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supportive families, and supportive health care professionals were their primary motivating factors. Barriers to self-management included a lack of knowledge and understanding of the condition lead to feelings of anxiety, as well as a lack of support from families and healthcare professionals. The following sections will address each sub-theme in detail.

Motivations for self-management

Desire for a healthy baby

The most significant motivating factor for self-management was the desire for a healthy baby. Pregnancy comes with specific concerns and anxieties, women felt anxious over the burden of self-management and the health of their baby. It was clear that many women wanted to *"do the right thing by the baby"* (15), to ensure they had a healthy start in life. Women expressed fear about the potential to pass their ill health onto the baby. Most often, the initial response to their diagnosis was concern for the baby; "*oh my God…you know is the baby gonna be OK?"* (16). Some women expressed feelings of responsibility for their condition, and their desire to minimise the effect on their baby's health; *"I knew I had brought it on myself by being overweight…I felt very responsible"* (15).

Supportive environment: family

When the familial support was positive, women were motivated to self-manage for reasons beyond the health of the baby: *"The biggest thing I can recommend is getting support. If you try to do it on your own, it's going to seem very confusing and tedious"* (17). When speaking about her partner one woman said *"he was very supportive in helping me get over things"* (18), due to her partner's support she was able to overcome her fear of blood glucose self-testing. Families and partners who

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adhered to the same self-management principles in their daily lifestyles helped women motivate themselves to follow dietary and exercise guidelines.

Supportive environment: health care professionals

Similarly, a supportive and constructive relationship with health care professionals proved to be a strong motivator for self-management, often minimising feelings of anxiety and uncertainty. Healthcare professionals who provided self-management advice gave women more autonomy over their pregnancy, *"the diabetes educator is really friendly…she did a demo in front of me, how to inject yourself"* (18). Women who understood their condition better were able to make positive choices about their self-management; *"I would limit my foods in order to control my blood sugar"* (19).

Barriers to self-management

Feeling anxious and overwhelmed

In this dataset, women often described feelings of anxiety and were overwhelmed by the prospect of managing their new condition. Women reported feelings of unhappiness or low mood while learning to self-manage, calling it "frustrating". Some women found trying to follow self-management guidelines created lot of anxiety: *"It creates a lot of anxiety. Like I am hungry but I don't want to eat now because I don't know if I harm the baby. But I feel hungry and could not eat"* (20). While it was often a lack of information that made women feel anxious, there was some evidence that being given *"too much information (about gestational diabetes)"* (15) also made women feel overwhelmed. Women were negative about self-management when they perceived that it had not gone to plan: *"When I took the higher dose [of insulin], I felt defeated…this is crazy, I don't think my body is ever going to be perfect"* (16).

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Lack of knowledge and understanding

A lack of knowledge and understanding often led to a lack of motivation to self-manage. Women who did not understand their condition often underestimated the seriousness of their diagnosis or heard anecdotally from other women that it was not *"a severe illness"* (21) thus preventing women from committing to self-management. Contrastingly, others felt as though it was *"the end of the world"* (17) leading to women feeling too overwhelmed to self-manage efficiently. Central to the lack of understanding was a perception of insufficient education from health care professionals; one woman said that she was given *"a list of food but no instructions"* meaning *"you really have to work it out for yourself"* (15).

Lack of support: family

This dataset observed traditional forms of support networks: women's families, friends, and their healthcare professionals. A key theme to emerge from the data was the negative effect of a lack of familial support, which made women feel isolated and alone. Without strong family support networks women either stopped complying with self-management or lost the motivation to self-manage. Concerning diet self-management women reported their families being unwilling to follow the same dietary restrictions because *"you can't expect everybody to change everything"* (15), or failing to understand the importance of the dietary guidelines. Some women expressed that adhering to the strict diabetes diet made them feel socially isolated.

Lack of support: health care professionals

A discordant relationship with healthcare professionals made women feel less motivated to self-manage*, "I love my OBGYN, but I feel they are always in a hurry. Like they don't have time to sit there and talk to you about what to do about it (GDM), but they are always in a hurry"* (17). One

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woman described her healthcare professional's attitude as *"here, you take the ball, and you run with it" (22),* but found that when her family were not informed about her self-management needs, it was difficult for her to stay on track. Women who wanted to self-manage reported either not being supported adequately by their healthcare professionals or subject to paternalistic responses: *"you have no idea what you're doing, I know best, here's what we're going to do"* (23). Some women had difficulty accessing information and understanding the self-management tasks, particularly when there were language barriers: *"It's a bit difficult because my English isn't very good…I have always been asking for an interpreter, but I have never got one"* (24). There were also concerns ranging from a lack of information (*"they didn't tell me what's the side effects for the baby"* (15)) to accusations of patronising healthcare professionals who *"talked down to [me]"* and *"didn't trust me"* (17).

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Discussion

Principle Findings

This review found that women with diabetes predominantly undertake self-management during pregnancy for the health of their baby. Support networks are crucial in alleviating the burden that comes with managing a chronic condition, particularly a new one, such as gestational diabetes. When these networks are less effective, women may feel overwhelmed by the self-management, which is compounded by a lack of knowledge and understanding.

One of the most notable findings from this review was the lack of research dedicated to managing chronic conditions in pregnancy. Diabetes was the only condition represented in the literature, but with only 16 papers eligible for inclusion, the qualitative evidence base around managing gestational or chronic diabetes in pregnancy is still thin. While the findings of this review are framed by the self-management challenges of diabetes, a number of generic activities are applicable across other chronic conditions. Self-management of diabetes includes self-testing (of blood glucose levels) and diet and exercise management, all of which are applicable more widely in pregnancy. However, it is clear there is a lack of specific research on other chronic conditions in pregnancy, such as hypertension. It is also worth noting that in the general population chronic conditions are self-managed due to the absence of a 'cure', but this is not strictly the case in pregnancy. While pre-existing chronic conditions in pregnancy do not have the possibility of a cure, some gestational chronic conditions do have that possibility. Gestational diabetes for example, is cured with delivery.

The results of this review have demonstrated that the motivating factors for pregnant women to self-manage are different to those motivating the general population; none more so than the desire

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to improve outcomes for the baby. As far as the authors are aware this paper is the first to use the burden of treatment theory as a theoretical lens to explore experiences of chronic disease management in pregnancy. Although this theory does not fully account for the specific challenges of pregnancy, as demonstrated by this review, it still broadly accounts for the actors and work involved in self-management in this context. Using this theory allowed for a deeper understanding of the burden of self-management and the work involved.

Interpretation

Acting for the Baby

May et al argue that the work of self-management in the general population is often done in the context of "holding together something larger and more complex" (14), alluding to everyday life. While this applies in pregnancy, the work of self-management is also done in the context of maintaining the health of the unborn baby. Previous literature has touched on women acting "not for the self but also for the unborn "other"'(25) and the burden of women having to be 'accountable' for the health of their baby alongside their own health cannot be underestimated. Current self-management models are not equipped to consider that particular burden of treatment.

Support Networks

Current self-management models emphasise that "self-management is manifested as both an individual and family construct" (26) and underscore the importance and necessity of social networks. The burden of treatment theory also acknowledges the importance of "members of [sick people's] social networks" to manage patient work. As both a barrier and a facilitator of self-management, family support is an important factor in continued adherence. While current self-

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management models do not accommodate pregnancy entirely, elements of them are applicable, the importance of social networks and the need for education regarding the diagnosis were evident in this review.

Strengths and Limitations

The research question evolved throughout the search as it became apparent that little data on self-management in pregnancy existed, and none outside of diabetes management. This review should act as a call for further work. Lived experiences are often over looked in self-management literature and by using qualitative literature we have been able to address some of these research gaps.

Limitations include that the coding framework is constrained by how the original researchers interpreted the interviews and their interactions with patients. Distinguishing between pre-existing chronic conditions and pregnancy acquired chronic conditions is important, but it was not a distinction that was reflected in the dataset; it is both a limitation of the included studies and this review. The burden of treatment theory does not perfectly map to pregnancy, however elements of this theory are helpful in understanding how and why patients interact with self-management. Many of the included papers specifically considered barriers to self-management, as opposed to facilitators, thus potentially creating a negative bias towards self-management.

Included studies took place across a variety of international settings, operating within different health systems, ranging from those dominated by private insurance-based systems to those with national healthcare systems aimed at universal healthcare coverage. Individualised medicine is intertwined with self-management; six of the included studies took place in the United States, where they operate within an insurance-based system that lends itself to an individualised approach to medicine. With the exception of Canada, New Zealand, and Singapore, none of the other countries included in this review have health systems based on almost near universal health

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care coverage. A patient's relationship with their healthcare system also affects their relationship with the acceptability of self-management and their efficacy undertaking it. As none of the included studies were conducted in the UK, it raises questions about how pregnancy self-management would operate within the NHS.

Conclusions

Clinical Implications

This review suggests there are several ways to improve adherence to self-management interventions in pregnancy, these interventions often involve self-monitoring and titrating medication. Health care professionals communicating the importance of self-management in terms that focus on the health of the baby, and educating women on the condition they have been diagnosed with, could have a positive impact on self-management adherence.

Research Implications

There is a need for research on the self-management of chronic conditions in pregnancy, particularly in chronic conditions other than diabetes. There are ongoing trials on self-management in pregnancy, predominantly in hypertension, but how self-management will fit into usual care remains to be seen. Remote forms of care are now more commonplace, in light of this, a better, qualitative understanding of the new burden of treatment self-management creates is needed.

Main Conclusions

This review shows that the primary motivating factor for women self-managing is the health of their baby. Their support networks and their understanding of the condition contribute to whether they self-manage effectively. The burden of treatment shifting to women requires further research, as patient work increases, feelings of anxiety can also increase. In this dataset, it was clear that

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some women found this treatment shift to be overwhelming but their anxiety largely stemmed from a lack of knowledge and understanding of the condition with which they had been diagnosed. Evidently, education is a barrier to pregnant women effectively self-managing, as when these issues are addressed women are willing and able to self-manage.

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Contribution to Authorship

BJ, KT, LH, RM and NR contributed to creating the research question and parameters of the review, including the search strategy. BJ conducted the search with assistance from NR. BJ and JK reviewed the titles and abstracts of identified articles and carried out the full text assessments. Disagreements on reviewed articles were resolved by consensus or discussion with KT. Data extraction and complete analysis was conducted by BJ and reviewed by KT, LH and RM. The write up was led by BJ, with edits provided by KT, LH and RM.

Competing interests

The authors declare no conflict of interest.

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Patient Consent

Not required

Data Sharing Statement

No original data were generated for this study

Public and Patient Involvement

Patients and the public were not involved in this systematic review. Since this is a qualitative systematic review of already published papers, and no participant recruitment took place, the involvement of patients and the public is not applicable.

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Tables

Table 1: Characteristics of Included Studies

Title	Year	Authors	Chronic Condition	Journal	Sample Size	Setting	Qualitative Method
Living with gestational diabetes in a rural community	2014	Abraham , K. Wilk, N.	Diabetes	The American Journal of Maternal and Child Nursing	10	USA	One on one interviews
Women's experiences of gestational diabetes self-management: A qualitative study	2013	Carolan, M.	Diabetes	Midwifery	15	Australia Diabetes clinic	One on one interviews, focus group
Women's experiences of factors that facilitate or inhibit gestational diabetes self-management	2012	Carolan, M. Gill, G. Steele, C.	Diabetes	BMC Pregnancy and Childbirth	15	Australia Diabetes clinic	One on one interviews, focus group
Experiences, perceptions and self-management of gestational diabetes in a group of overweight multiparous women	2014	Chavez-Courtois, M. Graham, C. Romeo-Perez, I. Sanchez-Miranda, G. Sanchez-Jimenez, B. Perichart-Perera, O.	Diabetes	Cien Saude Colet	5	Spain	One on one interviews
'Is gestational diabetes a severe illness?' exploring beliefs and self-care among women with gestational diabetes living in a rural area of the south east of China	2016	Ge, L. Wikby, K. Rask, M.	Diabetes	Australian Journal of Rural Health	17	China Obstetric clinics, wards	One on one interviews
The maternal experience of having diabetes in pregnancy	2011	Nolan, J. McCrone, S. Ilana, R. Chertok, A.	Diabetes	Journal of the American Academy of Nurse Practitioners	8	USA	Focus groups
Self-monitoring lifestyle behaviour in overweight and obese pregnant women: qualitative findings	2018	Sheih, C. Draucker, C.B	Overweight /obese	Journal of Clinical Nursing	13	USA Clinics (community health centre and health network)	One on one interviews
Socio-cultural aspects of self-management in gestational diabetes	2012	Wazqar, D. Evans, M.	Diabetes	Diabetes Nursing	12	Canada	Secondary analysis of one on one interviews
'I was tired of all the sticking and poking': Identifying barriers to diabetes self-care among low-income pregnant women	2015	Yee, L. McGuire, J. Taylor, S. Niznik, C. Simon, A.	Diabetes	Journal of Healthcare for the Poor and Underserved	10	USA Clinic	One on one interviews
Factors promoting diabetes self-care among low-income, minority pregnant women	2016	Yee, L. McGuire, J. Taylor, S. Niznik, C. Simon, A.	Diabetes	Journal of Perinatology	10	USA Diabetes clinic	One on one interviews
Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: a phenomenological research	2017	Youngwanichsetha, S. Phumdoung, S.	Diabetes	Journal of Clinical Nursing	30	Thailand Antenatal clinic, diabetes clinic, obstetric ward	One on one interviews
Self-management of gestational diabetes among Chinese migrants: A qualitative study	2018	Wah, Y. McGill, M. Wong, J. Ross, G. Harding, AJ. Krassa, I.	Diabetes	Women and Birth	18	Australia Antenatal clinic	One on one interviews
Ethnic Differences in Dietary Management of Gestational Diabetes Mellitus: A Mixed Methods Study Comparing Ethnic Chinese Immigrants and Australian Women	2019	Wan, C. Teede, H. Nankervis, A. Aroni, R.	Diabetes	Journal of the Academy of Nutrition and Dietetics	83	Australia Maternity services, tertiary hospitals	One on one interviews and diary entries

The experience of gestational diabetes for indigenous Māori women living in rural New Zealand: qualitative research informing the development of decolonising interventions	2018	Reid, J. Anderson, A. Cormack, D. Reid, P. Harwood, M.	Diabetes	BMC Pregnancy and Childbirth	10	New Zealand Primary care clinic	One on one interviews
"Diabetes Just Tends to Take Over Everything": Experiences of Support and Barriers to Diabetes Management for Pregnancy in Women With Type 1 Diabetes	2019	Singh, H. Ingersoll, K. Gonder-Frederick, L. Ritterband, L.	Diabetes	Diabetes Spectrum	15	USA Diabetes clinic	One on one interviews
Barriers to Gestational Diabetes Management and Preferred Interventions for Women With Gestational Diabetes in Singapore: Mixed Methods Study	2020	Hewage, S. Audimulam, J. Sullivan, E. Chi, C. Yew, T. W. Yoong, J.	Diabetes	JMIR Formative Research	15	Singapore Gestational Diabetes clinic	One on one interviews, in a mixed methods study also including a survey

Table 2: Quote Table: Barriers and Motivations for Self-management

Barriers to self-management		Motivations for self-management	
Lack of knowledge and understanding	"Actually, I didn't know anything about it...So, I...thought that if you were diabetic then you kind of get it. But then I didn't know something you can just develop during pregnancy as well. So, it was quite new to me." (27)	**Desire for healthy baby**	"I have a responsibility to care for my child…and to care for this baby inside of me…For me, I wasn't going to let anything get in the way…" (28)
Feeling anxious and overwhelmed	"With the fourth pregnancy, I had sort of lost control. I was living in the garage with my three children [and husband]. I was huge and six-months [along in my pregnancy] when I was diagnosed with depression….I didn't want to do anything. I'd just wake up [and] went through the motions." (29)	**Supportive environment: family**	"If it's not because of my husband, I couldn't have made it this far, like he would remind me, sometimes when I want to eat something he would just remind me not to, and like when we go for shopping he would buy the healthy stuff." (24)
Lack of support: family	"I need to cook for my husband and I can't make him eat the same thing…I need to follow him because he needs energy for his work…and I can't avoid using sauce, you know how on the menu it says you need to avoid sauces and stuff, I can't do that, my husband wouldn't want to eat according to that." (24)	**Supportive environment: HCP**	"The diabetes educator is really friendly…she explained things very like, in a very good way Yes, yes and – like, she did a demo in front of me, how to inject yourself. It was really scary first time. They told me everything." (19) "The first pregnancy I had a lady (educator) that was really excellent, very understanding, very approachable. She would listen to my concerns and we had a conversation rather than just a one-way flow of information. And so I had a very positive experience with her." (15)
Lack of support: HCP	"You are told that you have this and nobody spends time with you, it's kind of frustrating because you want to cry. You think that it's something you did wrong" (17) "According to my experience, what they (health professionals) discussed are all about the dietary habit of women here (in Australia), so if the suggestions could be more relevant to Chinese lifestyle, a lot of things that have been taught currently are not meaningful." (20)		

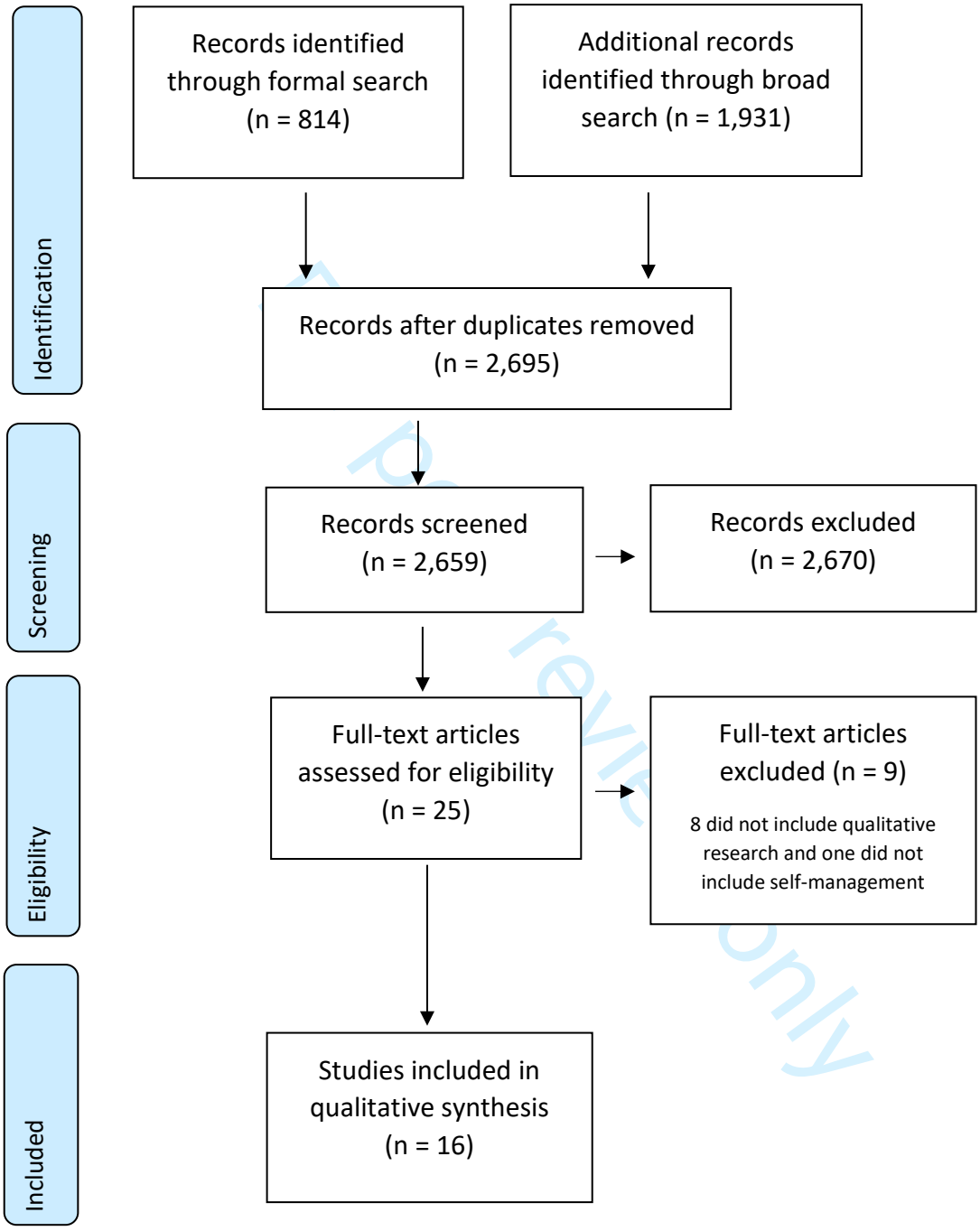
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Figures

Figure 1: PRISMA Flow Diagram

Legend: PRISMA Flow Diagram of study search strategy

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Appendix 1: Search Strategy

Two databases were searched, CINAHL and Pub Med. The search terms used to find articles in Pub Med (n=107) were MeSH terms 'pregnancy,' 'pregnant women', 'decision making', 'self-care', and 'pregnan* OR gestational OR maternal', and 'shared decision making OR self manag* OR self-monitor'. There was also a broad secondary search in Pub Med using the search term 'pregnancy self-management' which identified a further 1,931 articles for review. The search terms used for CINAHL (n=707) were 'pregnancy OR expectant mothers', 'pregnan* OR gestational OR maternal', 'self-care', and 'shared decision making OR self-manag* OR self-monitor'. This identified a total of 2,745 articles, which when controlled for duplicates left 2,695 for review. 25 articles were identified for full text review, of which 16 were selected for the systematic review.

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??? For those soon to register with PROSPERO, records may not be submitted between Thursday 1st April (5 pm) and Tuesday 6th (8 am) whilst staff are away and the university is closed for the Easter period. During this time you may still work on your record and save it for submission after Easter.

Systematic review

1. * Review title.

Give the title of the review in English

What are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?

2. Original language title.

For reviews in languages other than English, give the title in the original language. This will be displayed with the English language title.

3. * Anticipated or actual start date.

Give the date the systematic review started or is expected to start.

01/04/2019

4. * Anticipated completion date.

Give the date by which the review is expected to be completed.

31/12/2019

5. * Stage of review at time of this submission.

Tick the boxes to show which review tasks have been started and which have been completed. Update this field each time any amendments are made to a published record.

Reviews that have started data extraction (at the time of initial submission) are not eligible for inclusion in PROSPERO. If there is later evidence that incorrect status and/or completion date has been supplied, the published PROSPERO record will be marked as retracted.

This field uses answers to initial screening questions. It cannot be edited until after registration.

The review has not yet started: No

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Review stage	Started	Completed
Preliminary searches	Yes	No
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	Yes	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Provide any other relevant information about the stage of the review here.

6. * Named contact.

The named contact is the guarantor for the accuracy of the information in the register record. This may be any member of the review team.

Beth Jakubowski

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

Ms Jakubowski

7. * Named contact email.

Give the electronic email address of the named contact.

bethany.jakubowski@phc.ox.ac.uk

8. Named contact address

Give the full institutional/organisational postal address for the named contact.

9. Named contact phone number.

Give the telephone number for the named contact, including international dialling code.

01865 617961

10. * Organisational affiliation of the review.

Full title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

University of Oxford

Organisation web address:

https://www.phc.ox.ac.uk/

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NHS
National Institute for
Health Research

11. * Review team members and their organisational affiliations.

Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. **NOTE: email and country now MUST be entered for each person, unless you are amending a published record.**

Ms Bethany Jakubowski. University of Oxford
Ms Jaspreet Khaira. University of Oxford
Professor Richard McManus. University of Oxford
Dr Katherine Tucker. University of Oxford
Dr Lisa Hinton. University of Cambridge

12. * Funding sources/sponsors.

Details of the individuals, organizations, groups, companies or other legal entities who have funded or sponsored the review.

CLAHRC Oxford

Grant number(s)

State the funder, grant or award number and the date of award

13. * Conflicts of interest.

List actual or perceived conflicts of interest (financial or academic).

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country must be completed for each person, unless you are amending a published record.**

15. * Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

What are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

The search terms used to find articles in PubMed

- Pregnancy MeSH
- Pregnant women MeSH
- Pregnan* OR gestational OR maternal
- Decision making MeSH
- Self-care MeSH
- Shared decision making OR self manag* OR self-monitor

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The search terms used for CINAHL

- Pregnancy OR Expectant Mothers

- Pregnan* OR gestational OR maternal

- Self-care

- Shared decision making OR self-manag* OR self-monitor

There was also a broad secondary search using the search term 'pregnancy self-management' in PubMed

17. URL to search strategy.

Upload a file with your search strategy, or an example of a search strategy for a specific database, (including the keywords) in pdf or word format. In doing so you are consenting to the file being made publicly accessible. Or provide a URL or link to the strategy. Do NOT provide links to your search **results**.

https://www.crd.york.ac.uk/PROSPEROFILES/136681_STRATEGY_20200916.pdf

Alternatively, upload your search strategy to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

Do not make this file publicly available until the review is complete

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

This review is on the self-management of chronic conditions in pregnancy i.e. hypertension, diabetes, kidney disease. The scope of the review is to look at the feasibility and acceptability of women self-managing, and how they regard this new burden of treatment.

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

The population being studied is pregnant women with a chronic condition, there are no further restrictions on the study population.

Inclusion Criteria

1. Chronic condition: diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease, others to be reviewed as they come up

2. Pregnancy

3. Self-management: self-monitoring, self-weighing, self-care, others to be reviewed as they come up

4. Qualitative.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The

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preferred format includes details of both inclusion and exclusion criteria.

This review is looking at the self-management of chronic conditions in pregnancy. Self-management can be defined as self-monitoring (i.e. of blood glucose or blood pressure), or as action on the basis of self-monitoring (i.e. diet management, self-care, weight management).

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

Usual care if appropriate.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Qualitative studies are eligible for inclusion, including secondary analysis of previous qualitative research.

Inclusion Criteria

1. Chronic condition: diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease, others to be reviewed as they come up

2. Pregnancy

3. Self-management: self-monitoring, self-weighing, self-care

4. Qualitative

23. Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

Chronic conditions in pregnancy are under researched, so the aim of this review is to examine the current qualitative evidence of the behaviours and attitudes towards the self-management of chronic conditions in pregnancy. It is important to determine what, if any, self-management currently takes place and whether women find it acceptable and feasible to self-manage. Therefore the inclusion criteria for this review is for the participant to have a chronic condition (i.e. hypertension, diabetes) during pregnancy (this can be defined as gestational or chronic), and for some form of self-management to be taking place. This self-management can either be self-monitoring (i.e. of blood glucose or blood pressure), or action on the basis of self-monitoring (i.e. diet management, self-care, weight management).

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

The study aims to explore whether self-management of chronic conditions in pregnancy is acceptable and

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feasible to pregnant women. It also aims to explore current practice and the current evidence base of chronic disease management in pregnancy.

Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

None.

Measures of effect

Please specify the effect measure(s) for you additional outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

All of the studies identified in the search were screened independently by two reviewers (title and abstract) who then came together to agree on which articles should get a full text review. Both reviewers then independently completed the full text review and met again to agree on which articles should be included in the final review. Discrepancies were resolved by reviewing the definition of self-management and chronic condition as these were often the causes of uncertainty. The first author will then use a data extraction spreadsheet to list the key themes and findings from the papers. 10% of the included articles will be screened by another member of the study team, measuring the 10% of articles against the inclusion criteria and the COREQ quality assessment guidelines. The included studies will then be put into NVivo 12 and a thematic analysis, using the guidelines produced by Braun and Clarke (2006), will be carried out by the first author. A theoretical lens will be used to reflect the themes that emerge during data analysis.

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

Two researchers carried out the screening of articles, and met after the title and abstract screening to feed back results and decide which articles required a full text review. They met again following this screening to decide which articles should be included in the final review. To determine the quality of the studies, the COREQ guidelines will be used, which is a standard set of guidelines to determine the quality of qualitative work. The quality of the studies will be acknowledged in the write up, in order to be transparent about the

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quality of the work in this field to date.

28. * Strategy for data synthesis.

Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta-analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.

Using NVivo 12, an initial thematic analysis will be conducted by the first author to create a coding framework of themes and sub themes. A theoretical lens will be used to develop the coding framework further. Discrepancies will be discussed with the wider study team and feedback will be incorporated into the framework.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

Not currently planned

30. * Type and method of review.

Select the type of review, review method and health area from the lists below.

Type of review

Cost effectiveness
No

Diagnostic
No

Epidemiologic
No

Individual patient data (IPD) meta-analysis
No

Intervention
No

Meta-analysis
No

Methodology
No

Narrative synthesis
No

Network meta-analysis
No

Pre-clinical
No

Prevention
No

Prognostic

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No

Prospective meta-analysis (PMA)
No

Review of reviews
No

Service delivery
No

Synthesis of qualitative studies
Yes

Systematic review
Yes

Other
No

Health area of the review

Alcohol/substance misuse/abuse
No

Blood and immune system
No

Cancer
No

Cardiovascular
No

Care of the elderly
No

Child health
No

Complementary therapies
No

COVID-19
No

Crime and justice
No

Dental
No

Digestive system
No

Ear, nose and throat
No

Education
No

Endocrine and metabolic disorders
No

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Eye disorders
No

General interest
No

Genetics
No

Health inequalities/health equity
No

Infections and infestations
No

International development
No

Mental health and behavioural conditions
No

Musculoskeletal
No

Neurological
No

Nursing
No

Obstetrics and gynaecology
No

Oral health
No

Palliative care
No

Perioperative care
No

Physiotherapy
No

Pregnancy and childbirth
Yes

Public health (including social determinants of health)
No

Rehabilitation
No

Respiratory disorders
No

Service delivery
No

Skin disorders
No

Social care

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No

Surgery
No

Tropical Medicine
No

Urological
No

Wounds, injuries and accidents
No

Violence and abuse
No

31. Language.

Select each language individually to add it to the list below, use the bin icon to remove any added in error.
English

There is not an English language summary

32. * Country.

Select the country in which the review is being carried out. For multi-national collaborations select all the countries involved.

England

33. Other registration details.

Name any other organisation where the systematic review title or protocol is registered (e.g. Campbell, or The Joanna Briggs Institute) together with any unique identification number assigned by them. If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34. Reference and/or URL for published protocol.

If the protocol for this review is published provide details (authors, title and journal details, preferably in Vancouver format)

Add web link to the published protocol.

Or, upload your published protocol here in pdf format. Note that the upload will be publicly accessible.

No I do not make this file publicly available until the review is complete

Please note that the information required in the PROSPERO registration form must be completed in full even if access to a protocol is given.

35. Dissemination plans.

Do you intend to publish the review on completion?

Yes

Give brief details of plans for communicating review findings.?

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We will aim to publish the review in a high impact scientific journal.

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords help PROSPERO users find your review (keywords do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

Pregnancy; maternal health; chronic conditions; self-management

37. Details of any existing review of the same topic by the same authors.

If you are registering an update of an existing review give details of the earlier versions and include a full bibliographic reference, if available.

38. * Current review status.

Update review status when the review is completed and when it is published.New registrations must be ongoing so this field is not editable for initial submission.
Please provide anticipated publication date

Review_Ongoing

39. Any additional information.

Provide any other information relevant to the registration of this review.

40. Details of final report/publication(s) or preprints if available.

Leave empty until publication details are available OR you have a link to a preprint (NOTE: this field is not editable for initial submission). List authors, title and journal details preferably in Vancouver format.

Give the link to the published review or preprint.

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Quality Appraisal Checklist	Personal Characteristics				Relationship with Participants			Theoretical Framework	Participant Selection				Setting			Data collection						
Title	Interviewer/Facilitator	Credentials	Occupation	Experience and training	Relationship established	Participant knowledge of the interviewer/interview	Interviewer characteristics	Methodological orientation and theory	Sampling	Method of approach	Sample size	Non-participation	Setting of data selection	Presence of non-participants	Description of sample	Interview guide	Repeat interviews	Audio/visual recording	Field notes	Duration	Data saturation	Transcripts returned
Self-monitoring lifestyle behaviour in overweight and obese pregnant women: qualitative findings	First author	Author 1: DNSc, MPH, RNC-OB, FAAN. Author 2: PhD, RN, FAAN	Author 1: Associate Professor. Author 2: Researcher	First author: Expert in prenatal health promotion and content analysis. Second author: PhD. Doctoral student carrying out analysis: undergraduate degree in exercise science, experience in a home visitation programme for low-income pregnant women	First author called or e-mailed each woman 3 - 7 days before a scheduled interview	Not mentioned - assumed through recruitment	No bias reported	Self-monitoring theory, behavioural and lifestyle change theories	Purposive	Interview explained to trial participants and informed consent sought	13	22 women enrolled in the intervention, 9 refused interview	10 interviewed in clinic private rooms, 3 over phone	No	Mean age 29.5 years, majority obese, majority were African American (77%), unmarried (69%), previous pregnancy (77%), third trimester (69%)	Yes	No	Audio	No	30 minutes	Not discussed	No
"I was tired of all the sticking and poking": Identifying Barriers to Diabetes Self-Care Among Low-Income Pregnant Women	Author 1 and 2	Author 1: MD, MPH, Author 2: MD, MPH, Author 3: MPH, Author 4: APN, CDE, Author 5: MD, MPH	Author 1: Fellow in Maternal-Fetal Medicine. Author 2: Resident in OBGYN at Northwestern. Author 3: Senior Research Coordinator at Northwestern. Author 4: Advance Practice Nurse in Maternal-Fetal Medicine. Author 5: Professor of OBGYN at Northwestern	Extensive expertise in maternal-fetal medicine and research	Not mentioned	Were identified because they had GDM so knew why the research was being conducted	No bias reported	Cognitive load theory	Purposive	At a clinic serving public-assisted low-income women. Informed consent sought.	10	46 women identified, 22 ineligible, 12 declined participation	Conducted in private	No	Mean age 28.5, African American (4), Latina (4), Native American (1), Asian (1). Most educated to high school degree level, most multiparous, all diabetic	Yes	Three interviews per person	Audio	No	30-60 minutes	Reached at 29 interviews	No
The maternal experience of having diabetes in pregnancy	Not reported by coder/analyser was present for all interviews and focus groups	Author 1: MD, Author 2: PhD, RN, PMHCNS-BC, Author 3: PhD, RN, IBCLC	Not reported	Extensive expertise judging by credentials	Not mentioned	Recruitment flyers provided some knowledge of subject matter	No bias reported	Phenomenology	Convenience	Flyers distributed in healthcare provider offices	8	Not reported	Over the phone or in person	Yes, additional investigator to take field notes	Average age 35, mostly highly educated, no demographics reported	Yes	No - these were focus groups	Audio	Yes	Not reported	Not discussed	Some were to check participants views were represented correctly
Women's experiences of gestational diabetes self-management: a qualitative study	Not reported	Author 1: PhD	Associate Professor Midwifery	Not reported but has a PhD in a related field	Approached by researcher prior to interview to confirm interest	Recruited from diabetes clinic fitting criteria	No bias reported	Phenomenology	Purposive (blinded for review)	Women invited to participate if fitted criteria, approached by researcher if Diabetes Coordinator confirmed their interest	15	Not reported	10 phone interviews, one in interviewee's home, one focus group at the hospital clinic	No	Variety of ethnic backgrounds, 23 - 40, majority high school educated, majority primiparous	Not reported	No	Audio	No	Not reported	Not discussed	No
Factors promoting diabetes self-care among low-income, minority pregnant women	Author 1 and 2	Author 1: MD, MPH, Author 2: MD, MPH, Author 3: MPH, Author 4: APN, CDE, Author 5: MD, MPH	Author 1: Fellow in Maternal-Fetal Medicine. Author 2: Resident in OBGYN at Northwestern. Author 3: Senior Research Coordinator at Northwestern. Author 4: Advance Practice Nurse in Maternal-Fetal Medicine. Author 5: Professor of OBGYN at Northwestern	Members of the research team reported to have prior qualitative research experience and extensive interviewing training	Not mentioned	Were identified because they had GDM and at their first clinic visit so knew why the research was being conducted	No bias reported	Cognitive load theory	Purposive	Recruited at their first clinic visit for diabetes care	10	46 women identified, 22 ineligible, 12 declined participation	Conducted in private	No	Mean age 28.5, African American (4), Latina (4), Native American (1), Asian (1). Most educated to high school degree level, most multiparous, all diabetic	Yes, provided in article	Three interviews per person	Audio	No	30-60 minutes	Reached at 29 interviews	No
Socio-cultural aspects of self-management in gestational diabetes	Author 1	Author 1: PhD student. Author 2: Assistant Professor	Author 1: PhD student. Author 2: Assistant Professor Nursing	PhD student	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Phenomenology	Purposive	Women being treated for GD were recruited	12	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Aged between 23 - 38, most worked outside the home, most were married, mix of first pregnancy and second pregnancies	Yes	Two interviews conducted per person	Audio	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis
Women's experiences of factors that facilitate or inhibit gestational diabetes self-management	Author 1	Not reported	Not reported	Not reported	Access to women facilitated by diabetes co-ordinator, no relationship pre-established with researcher	Recruited through diabetes clinic and by diabetes co-ordinator so would have been told what the study was about	No bias reported	Interpretative phenomenology	Purposive	Diabetes coordinator approached women, participants were recruited after they attended GDM education and had at least 3 weeks self management of condition	15	30 women met inclusion criteria, 20 agreed to participate, 15 available on day of interview	Focus group in private room in clinic, phone interviews, face to face interview at one woman's home	No	Age range 23 -40, education level lower than population average, mix of ethnic backgrounds, majority of women primiparous	Not reported	No	Audio	No	Not reported	Not mentioned	No
Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: a phenomenological research	Not reported	Author 1: PhD, RN, Assistant Professor of Nursing. Author 2: PhD, RN, Associate Professor of Nursing	Author 1: Assistant Professor. Author 2: Associate Professor	Not reported	Prolonged engagement was carried out by researchers to build a rapport with participants before interviews	Participants approached by staff nurses so knew about what the study was about prior to taking part	No bias reported	Phenomenology	Purposive	Approached at antenatal care clinics and informed consent obtained	30	Not reported	Interviewed at the antenatal clinic, diabetes clinic or obstetric ward	No	The mean age of participants 32.5 years, 17 women were Buddhists, 13 women were Muslims. Majority multiparous and employed.	Yes	Two interviews conducted per person	Audio	No	30 - 45 minutes	Data were collected until saturation (after six months)	No
Living with gestational diabetes in a rural community	Not reported	Author 1: DNP, APRN-C. Author 2: DNS, WHNP-C	Author 1: Visiting Assistant Professor of Nursing. Author 2: Visiting Assistant Professor	Not reported	Established through recruitment	Had diagnosis of the disease which was being explored in interviews, and recruited via study posters	No bias reported	Phenomenology	Purposive and snowball	Information about study posted in obstetric and health care provider officers in New York State	10	Not reported	Interviewed in person or at phone	No	All women Caucasian between 25-49. All had history of GDM.	Yes	Follow up interviews conducted with four participants	Audio	No	15 - 30 minutes, second interviews 15 - 30 minutes	Checked by sharing core themes with participants	Transcripts of first interview provided to participants doing a second interview for verification
"Is gestational diabetes a severe illness?" Exploring beliefs and self-care behaviour among women with gestational diabetes living in a rural area of the south east of China	Author 1	Author 1: M, Msc. Author 2: RN, PhD. Author 3: RN, PhD	Author 1 a teacher of maternal care and bilingual	Not reported	Not mentioned	Interviewer did not know participants prior to study, was not part of the clinical care team.	No bias reported	No specific theoretical framework reported	Purposive	Women purposefully recruited from obstetric clinics and wards at a specific hospital.	17	Not reported	Interviewed in a room at the hospital	No	27.5 median age, six women high educational level, five middle educational level, six low educational level	Yes and provided in paper	No	Audio	No	40 - 60 minutes	Not mentioned	No
Experiences, perceptions and self management of gestational diabetes in a group of overweight multiparous women	Not reported	Not reported	Not reported	Interviews conducted by experts in anthropology and social psychology	Established through recruitment and screening process	Women were given a health questionnaire prior to recruitment	No bias reported	Phenomenology	Purposive	Women recruited through department of diabetes, given questionnaire and selected based on the questionnaire results	5	Not reported	Not reported	No	Reported in table	Yes	No	Audio	Yes	90 minutes	Not mentioned	No
Ethnic Differences in Dietary Management of Gestational Diabetes Mellitus: A Mixed Methods Study Comparing Ethnic Chinese Immigrants and Australian Women	Not reported	Author 1: MN, APD Author 2: PhD Author 3: MD Author 4: PhD	Not reported	Not reported	No	Were identified because they had GDM so knew why the research was being conducted	No bias reported	Symbolic interactionist framework and grounded theory	Purposive	Women recruited through GDM clinics	83	11 declined to participate and reasons were given	Not reported but some on the phone (others in person)	No	Reported in table	Yes	No	Audio	Yes	45 - 90 minutes	Not mentioned but large sample size so assumed there was saturation	No
Self-management of gestational diabetes among Chinese migrants: A qualitative study	Chief Investigator	Not reported	Faculties reported	CI was fluent in English, Mandarin and Cantonese	Participant's had interviewer's contact details	Interested participants were provided with written information that explained the background of the study and provided the contact details of the researchers. All participants were recruited after they had received diabetes education from a diabetes educator and had at least two weeks of self-management experience.	No bias reported	Thematic analysis	Purposive	Between August and October 2016, participants were purposively recruited from the antenatal clinic at the RPAH in the Inner West Region of Sydney, Australia. Bilingual flyers about the study were distributed at the clinic. Interested participants were provided with written information that explained the background of the study and provided the contact details of the researchers.	18	Not reported	The interviews were conducted by the chief investigator in either a staff meeting room or a waiting room at the hospital, according to the participant's preference.	Yes, participant's partner present in two interviews	Reported in table: The mean age of participants was 34 years (SD = 3.6; range 29–41). The majority of participants came from China, Hong Kong or Taiwan and held a university degree	Yes	No	Audio	No	Not reported	Not reported	No
The experience of gestational diabetes for indigenous Māori women living in rural New Zealand: qualitative research informing the development of decolonising interventions	Not reported	Not reported	Faculties reported	Not reported	Lead researcher spoke to participants ahead of the interviewer	The contact details of all consenting participants were then provided to the lead researcher, who contacted participants via phone or home visits (when no phone service was available). The lead investigator provided detailed information about the research, answered any questions from women and obtained written informed consent and confidentiality agreements (including accessing their diabetes-related health data from the clinic) for each participant.	No bias reported	Kaupapa Maoris Methodology	Purposive - a sub sample of a larger study	The ten participants advised that they either had a clinical history of GDM (n = 8) or had been exposed to diabetes in utero (n= 2). Eligible participants were recruited via the local primary care clinic between September and December of 2015.	10	Reported in the larger study (6 withdrew)	Interviews undertaken in participants' homes or the health centre, or a mutually agreed venue.	No	10 Māori women aged between 30 and 69 years. Outlined in full table, participants had diverse pregnancy experiences and current diabetes-related health conditions	Yes	No	Audio	No	Not reported	Not discussed	Participants were invited to review, edit or change their transcript prior to data analysis.
"Diabetes Just Tends to Take Over Everything": Experiences of Support and Barriers to Diabetes Management for Pregnancy in Women With Type 1 Diabetes	Not reported	Not reported	Not reported	The main interviewer had a background in health psychology and skills in motivational interviewing.	Potential participants could indicate interest by contacting the investigator. Study procedures and the consent process reviewed with them.	Potential participants were informed that the purpose of the study was to identify barriers to and supports for blood glucose management around pregnancy	No bias reported	Thematic analysis	Purposive	Participants were recruited using advertisement flyers that were sent for distribution to diabetes specific clinics around the United States. Adult U.S. women (≥18 years of age) with type 1 diabetes who were fluent in English and were either pregnant, planning pregnancy for the immediate future, or had experienced a previous pregnancy were invited to participate in a phone interview.	15	27 indicated interest	Decided to conduct the interviews over the phone unless participants were local to the study site and wished to come to the study office	No	All participants were married, with a mean age of 34 years and a mean duration of type 1 diabetes of 21 years. Ten participants used a continuous glucose monitoring (CGM) system, and 13 used an insulin pump to manage their diabetes. Ten women had previously been pregnant, three were pregnant at the time of the interview, and five were trying for or contemplating pregnancy in the immediate future (within 0–4 months from the time of their interview).	Yes	No	Audio	Yes - notes and reflections recorded by interviewer	Between 30 and 50 minutes	To achieve data saturation, 10–15 people were recruited, with the possibility of conducting additional interviews, if necessary	No
Barriers to Gestational Diabetes Management and Preferred Interventions for Women With Gestational Diabetes in Singapore: Mixed Methods Study	First and second author	Author 1, PhD, Author 2 MPH, Author 3 MPH, Author 4 MBBS, MD, Author 5, MBBS, Author 6, PhD	Faculties reported	Not reported	Established through recruitment	Recruited from diabetes clinic fitting criteria, participants had previously participated in a survey on the same subject.	No bias reported	Thematic analysis	Purposive	Potential participants for the survey (this is a mixed methods study and interview participants were recruited from survey participants) were approached in the waiting room of the GDM clinic. The study team refrained from approaching women who seemed tired or anxious. After getting informed consent, women were given a self-administered survey. For pregnant women who were willing to participate in prenatal interviews, sessions were scheduled based on their convenience. In the second phase, to increase survey completion, the women were given an incentive of $25 (US $17.5). In the first phase, 50 patients participated in the survey.	15	Of the 50 patients who completed the survey, 35 gave verbal consent to participate in the interviews and 15 participated in the interviews	Most interviews were conducted at the health care facility or at the participants' homes, in a private space conducive for the participant to share thoughts effectively.	No	The women who participated in interviews had a demographic profile similar to that of the survey questionnaire participants. Half of the 15 participants in the age group of 31 to 35 years were primiparous or had a family history of T2DM, and most of them were Chinese or Indian, employed full time, with a degree, and attending a private prenatal clinic.	Yes	No	Audio	No	Not reported	Reported at 15 interviews	No

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Data analysis					Reporting					Reasons
Number of data coders	Description of coding tree	Derivation of themes	Software	Participant checking	Quotations presented	Data and findings consistent	Clarity of major themes	Clarify of minor themes	Quality Assessment	
2	Yes	Themes derived from data	N/A	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	Medium	Small sample size, short interviews, however themes were clear and recruitment process was transparent
2	Yes	Themes derived from data	ATLAS.ti 6	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	High	Despite small sample size, multiple interviews allowed for a richer dataset.
1 but checked informally by members of research team	No	Themes derived from data and then refined based on existing evidence of topic	N/A	Yes	Some reported, no quote table	Yes	Clear	Clear	Medium	Small sample size, even though focus groups tend to be better with a smaller sample size, this was still quite low and does not allow for a rich data set and exploration of themes
1	No	Themes derived from data	N/A	No	Yes with a quote table	Yes	Clear	Clear	High	Good sample size, clear discussion of themes and recruitment
2	Yes	Themes derived from data	ATLAS.ti 6	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	High	Despite small sample size, multiple interviews allowed for a richer dataset.
1	Yes	Themes derived from data	Not mentioned	Not mentioned - secondary analysis	No quote table, in text quotes included	Yes	Clear	Clear	Medium	A good sample size for a secondary analysis, but more information on primary data collection needed for transparency
1	Yes	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	High	Good sample size and mixed qualitative methods, leading to an apparently rich dataset. Paper was clear.
1	Yes	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	High	Good sample size , rich dataset and exploration of themes. Clear paper.
Not reported	No	Themes derived from data	Not mentioned	Yes participants were informed of core themes for member checking and saturation verification	No quote table, in text quotes included	Yes	Clear	Clear	Medium	Small sample size, but the discussion of themes was nuanced and rich. The interviews were also reported as being quite short.
Not reported	No	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	Medium	Good sample size with rich, longer form interviews. Clear paper and recruitment process clearly explained.
Not reported	No	Themes derived from data	ATLAS.ti V.5	No	No quote table, in text quotes included	Yes	Clear	Clear	Low	Very small sample size, potentially too low to make any assumptions about experiences of self-management
At least 2	No	Themes derived from data	Nvivo 11	No	Yes and quote tables included	Yes	Clear	Clear	High	Very large sample size, very rich dataset, clear paper and all processes communicated well
2	No	Themes derived from data	Not mentioned	No	Yes, embedded in article	Yes	Clear	Clear	High	Good sample size, clear paper and clear recruitment and analysis processes described
The lead researcher independently coded the data, and discussed the codes and themes with two experiences kaupapa Māori researchers, who had access to the interview data	No	Themes derived from data	Not mentioned	Participants were invited to review, edit or change their transcript prior to data analysis.	Yes, embedded in article	Yes	Clear	Clear	Medium	Small sample size, but an exploratory study and acknowledged it was a sub set of women who are rarely interviewed about their experiences.
Each transcript reviewed and coded independently by two researchers	Yes	Themes derived from data	Not mentioned	No	Yes, embedded in article	Yes	Clear	Clear	High	Good sample size, clear description of analysis process and themes very clear and supplemented with quotes
1	Yes	Themes derived from data	Atlas Ti	No - but transcripts checked by an author for accuracy	Yes but quote tables not included	Yes	Clear	Clear	Medium	Based on the qualitative interviews alone the sample size is small. However, there was a rich discussion of themes and analysis was rigorous.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3, 4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5, 6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5, 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7, 8, 10, 25, 26
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7, 8, 10, 25, 26
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7, 8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	N/A
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9, 10, 11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10 - 14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15, 16, 17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17, 18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18, 19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19, 20

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BMJ Open

Is self-management a burden? What are the experiences of women self-managing chronic conditions during pregnancy? A systematic review.

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Is self-management a burden? What are the experiences of women self-managing chronic conditions during pregnancy? A systematic review.

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Abstract

Objective

This systematic review examines the qualitative literature on women's experiences of self-managing chronic conditions in pregnancy.

Design

Systematic review of qualitative literature. Searches were performed in PubMed and CINAHL from inception to February 2021. Critical interpretive synthesis informed the coding framework and the analysis of the data. The Burden of Treatment theory emerged during the initial analysis as having the most synergy with the included literature, themes were refined to consider key concepts from this theory.

Participants

Pregnant women who are self-managing a chronic condition.

Results

A total of 2,695 articles were screened and 25 were reviewed in detail. All 16 included studies concerned diabetes self-management in pregnancy. Common themes coalesced around motivations for, and barriers to, self-management. Women self-managed primarily for the health of their baby. Barriers identified were anxiety, lack of understanding, and a lack of support from families and healthcare professionals.

Conclusions

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Pregnant women have different motivating factors for self-management than the general population and further research on a range of self-management of chronic conditions in pregnancy is needed.

Prospero Registration Number

CRD42019136681

Keywords: Systematic reviews, pregnancy, diabetes in pregnancy, chronic conditions, qualitative research, antenatal care, self-management, self-care, self-efficacy

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Article Summary

Short Title: Is self-management a burden?

Strengths and limitations of this study

- Pregnant women's experiences of self-managing during pregnancy is under researched compared to the general population. Lived experiences are often over looked in self-management literature and by using qualitative literature we have been able to address some of these research gaps.
- Limitations include that the coding framework is constrained by how the original researchers interpreted the interviews and their interactions with patients.
- Distinguishing between pre-existing chronic conditions and pregnancy acquired chronic conditions is important, but it was not a distinction that was reflected in the dataset; it is both a limitation of the included studies and this review.
- The burden of treatment theory does not perfectly map to pregnancy, however elements of this theory are helpful in understanding how and why patients interact with self-management.

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Introduction

Chronic diseases that manifest during pregnancy, or exist pre-pregnancy, can have a lasting impact on the health of the mother (1). Numerous conditions can affect women during pregnancy, some confined to that time (for example pre-eclampsia or gestational diabetes) and others that are longer lasting (chronic hypertension and diabetes). These conditions may lead to harm for both the mother and the baby; hypertensive disorders, for example, are one of the leading causes of maternal mortality globally (2).

Self-management

Self-management has been shown to support and improve the management of chronic conditions in the general population and there is a growing body of research to suggest self-management of chronic conditions in the pregnant population is also effective (3) (4) (5). Self-management interventions in pregnancy tend to focus on self-monitoring (blood pressure, blood glucose etc.) and in some cases titrating medication (insulin). Self-management following a hypertensive pregnancy has been explored post-partum, the evidence to date suggests not only is it feasible, but it may result in better diastolic blood pressure control at six months post-partum (4). In the antenatal period, the OPTIMUM-BP pilot trial suggested that self-monitoring of blood pressure was both feasible and acceptable to women with chronic hypertension (5). Self-management of diabetes in pregnancy has also been explored. In pregnant women with type 1 and type 2 diabetes, self-monitoring of blood glucose is common (6) and self-management programmes in women with gestational diabetes are also becoming more commonplace (3) (7). In the general population, self-management has been explored in numerous chronic conditions; in particular hypertension and diabetes, where there is a large body of research around understanding the patient experience (8) (9) (10). Much of this research has demonstrated patient's confidence and ability to self-manage

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(sometimes referred to as self-efficacy) (8) and suggests that self-management strategies improve health outcomes for those with chronic conditions (11). These conditions also affect women during pregnancy, but thus far this population are under-represented in this field of research. Moreover, research on self-management in the general population may not be wholly applicable to the pregnant population. Pregnancy presents a different set of motivations for self-management, not accounted for in current self-management frameworks which often assume long-term self-management, whereas with pregnancy the period of management is temporarily bounded even if it continues post-partum and considerations are for the mother and baby dyad.

Methods

We took a critical interpretative synthesis (CIS) approach (12). This allowed us to iteratively develop the research question and the inclusion criteria as the search progressed and the lack of research on the self-management of chronic conditions in pregnancy became apparent. In light of a lack of data, it was necessary to limit the scope of the review to the following research question: "what are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?"

We registered this review in PROSPERO (CRD42019136681, see Appendix 1, Supplementary File). Searches were performed in PubMed and CINAHL [EBSCOHost] (1982-present) from inception to February 2021 (see Appendix 2, Supplementary File).

Potentially eligible studies were screened by two independent reviewers against the following inclusion criteria:

1. Chronic condition: including but not exclusively diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease.
2. Pregnancy.

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3. Self-management: including but not exclusively self-monitoring, self-weighing, self-care.
4. Qualitative methods.

This review used a broad definition of self-management, "*the ability of an individual, in conjunction with family, community and health care professionals, to manage symptoms, treatments and lifestyle changes*" (13). We expanded the search criteria retrospectively to include risk factors for chronic conditions, in order to capture as much data as possible. The authors felt that qualitative literature was best placed to explore the lived experiences of people self-managing during pregnancy. Papers were excluded if the reviewers concluded that commonly recognised qualitative methods were not used, i.e. interviews, focus groups, or ethnographic observations.

Two reviewers (BJ & JK) independently reviewed the titles and abstracts of identified articles, a full text assessment of the relevant papers by both reviewers followed. The reference lists of included studies were screened but found no further articles to include. Disagreements on articles were resolved by consensus or discussion with a third reviewer (KT). Data extraction was conducted by BJ and checked by the study team. It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

Data Analysis

CIS also informed the analysis of the data (12). Analysis began with a detailed examination of the included papers, identifying themes and developing an understanding of the material. Dixon-Woods et al emphasise what distinguishes a systematic review using a CIS approach is *"the aim of being critical*" and questioning "*taken-for-granted-assumptions*" (12) meaning that the coding framework for this review was developed to critique the idea that general population self-management research was applicable to pregnancy. During the initial analysis, elements of the Burden of Treatment theory, developed for chronic condition management in the general

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population, were identified in the included literature. The coding framework was revised considering this theory's key concepts, such as the effect of patient work (e.g. self-monitoring, self-management, lifestyle changes) on self-management efficacy and the role of support networks, allowing for a more nuanced interpretation of the woman's experience of self-management. Data extraction was conducted by BJ using NVivo 12 Pro, and a coding framework was developed by BJ and agreed upon by the study team (LH, KT, RM).

The Burden of Treatment

May et al have developed a theory explaining the relationship between sick people and their healthcare providers in the context of chronic disease. In the absence of a possible cure, patients must engage in 'routine work' to manage the disease. May explains that patient work in the context of the Burden of Treatment theory includes symptom monitoring, treatment management, and lifestyle changes while "*maintaining* [these tasks] *alongside the demands of other aspects of everyday life*" (14). Transferring the burden of treatment to the patient prompts a change in the patient-clinician relationship. Within pregnancy this could theoretically provide women with more agency, as well as more responsibility.

Elements from the Burden of Treatment theory were identified in the first interpretation of the data, reviewing the themes that already existed in the included papers; therefore, the coding framework was iteratively mapped to two broad themes reflecting elements of this theory. These themes were grouped around motivations for self-management and barriers to self-management. As the coding progressed, more elements of the theory were incorporated into the analysis. May claims "*people and members of their social networks are expected to identify, understand and explain the diverse tasks that make up their work*" (14), and goes on to call this process "*sense*

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making". This idea was reflected in the themes concerning support from health care professionals and family, the impact of a lack of support, and women's anxiety and understanding of their diagnosis.

Results

Searches in PubMed and CINAHL identified 2,745 articles, which when controlled for duplicates left 2,695 for screening (see Appendix 2 and Figure 1 for further details, Supplementary File). The majority of articles were excluded as they did not include qualitative research or self-management, or the study population was not pregnant women. 25 articles were identified by the two reviewers for full text review, of which 16 fulfilled the inclusion criteria (see Table 1). Data quality checks using the COREQ guidelines were carried out (see Appendix 3, Supplementary File). Fifteen of the included studies were of medium to high quality. One was considered low quality because of the small sample size.

Included studies were published between 2011 and 2020 and all focused on diabetes in pregnancy. Six were conducted in the United States, four in Australia, and the remaining studies were carried out in Canada, Spain, Thailand, China, Singapore and New Zealand. The studies recruited from a range of settings: gestational diabetes clinics, antenatal clinics supporting low-income women, health care provider's offices, day care centres, and hospital based obstetric clinics. Fifteen of the studies used one-to-one interviews and one used focus groups for primary data collection. Two studies supplemented the interviews with focus groups, one supplemented with additional diary entries. One study was a mixed methods study, including one-to-one interviews alongside survey data. The study population for all included studies was pregnant women, although one study had a mixed population of women who were currently pregnant, women planning on getting pregnant, and women who had previously been pregnant. Only papers concerning diabetes self-management

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in pregnancy fulfilled the inclusion criteria, leading to the results focusing on the challenges of self-management in this disease.

The two broad themes that emerged from the data were motivations for and, conversely, barriers to self-management (see Table 2). Under the former, women express that having a healthy baby, supportive families, and supportive health care professionals were their primary motivating factors. Barriers to self-management included a lack of knowledge and understanding of the condition lead to feelings of anxiety, as well as a lack of support from families and healthcare professionals. The following sections will address each sub-theme in detail.

Motivations for self-management

Desire for a healthy baby

The most significant motivating factor for self-management was the desire for a healthy baby. Pregnancy comes with specific concerns and anxieties, women felt anxious over the burden of self-management and the health of their baby. It was clear that many women wanted to *"do the right thing by the baby"* (15), to ensure they had a healthy start in life. Women expressed fear about the potential to pass their ill health onto the baby. Most often, the initial response to their diagnosis was concern for the baby; "*oh my God…you know is the baby gonna be OK?"* (16). Some women expressed feelings of responsibility for their condition, and their desire to minimise the effect on their baby's health; *"I knew I had brought it on myself by being overweight…I felt very responsible"* (15).

Supportive environment: family

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When the familial support was positive, women were motivated to self-manage for reasons beyond the health of the baby: *"The biggest thing I can recommend is getting support. If you try to do it on your own, it's going to seem very confusing and tedious"* (17). When speaking about her partner one woman said *"he was very supportive in helping me get over things"* (18), due to her partner's support she was able to overcome her fear of blood glucose self-testing. Families and partners who adhered to the same self-management principles in their daily lifestyles helped women motivate themselves to follow dietary and exercise guidelines.

Supportive environment: health care professionals

Similarly, a supportive and constructive relationship with health care professionals proved to be a strong motivator for self-management, often minimising feelings of anxiety and uncertainty. Healthcare professionals who provided self-management advice gave women more autonomy over their pregnancy, *"the diabetes educator is really friendly…she did a demo in front of me, how to inject yourself"* (18). Women who understood their condition better were able to make positive choices about their self-management; *"I would limit my foods in order to control my blood sugar"* (19).

Barriers to self-management

Feeling anxious and overwhelmed

In this dataset, women often described feelings of anxiety and were overwhelmed by the prospect of managing their new condition. Women reported feelings of unhappiness or low mood while learning to self-manage, calling it "*frustrating*". Some women found trying to follow self-management guidelines created lot of anxiety: *"It creates a lot of anxiety. Like I am hungry but I don't want to eat now because I don't know if I harm the baby. But I feel hungry and could not eat"*

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(20). While it was often a lack of information that made women feel anxious, there was some evidence that being given *"too much information (about gestational diabetes)"* (15) also made women feel overwhelmed. Women were negative about self-management when they perceived that it had not gone to plan: *"When I took the higher dose [of insulin], I felt defeated…this is crazy, I don't think my body is ever going to be perfect"* (16).

Lack of knowledge and understanding

A lack of knowledge and understanding often led to a lack of motivation to self-manage. Women who did not understand their condition often underestimated the seriousness of their diagnosis or heard anecdotally from other women that it was not *"a severe illness"* (21) thus preventing women from committing to self-management. Contrastingly, others felt as though it was *"the end of the world"* (17) leading to women feeling too overwhelmed to self-manage efficiently. Central to the lack of understanding was a perception of insufficient education from health care professionals; one woman said that she was given *"a list of food but no instructions"* meaning *"you really have to work it out for yourself"* (15).

Lack of support: family

This dataset observed traditional forms of support networks: women's families, friends, and their healthcare professionals. A key theme to emerge from the data was the negative effect of a lack of familial support, which made women feel isolated and alone. Without strong family support networks women either stopped complying with self-management or lost the motivation to self-manage. Concerning diet self-management women reported their families being unwilling to follow the same dietary restrictions because *"you can't expect everybody to change everything"* (15), or failing to understand the importance of the dietary guidelines. Some women expressed that adhering to the strict diabetes diet made them feel socially isolated.

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Lack of support: health care professionals

A discordant relationship with healthcare professionals made women feel less motivated to self-manage, *"I love my OBGYN, but I feel they are always in a hurry. Like they don't have time to sit there and talk to you about what to do about it (GDM), but they are always in a hurry"* (17). One woman described her healthcare professional's attitude as *"here, you take the ball, and you run with it" (22),* but found that when her family were not informed about her self-management needs, it was difficult for her to stay on track. Women who wanted to self-manage reported either not being supported adequately by their healthcare professionals or subject to paternalistic responses: *"you have no idea what you're doing, I know best, here's what we're going to do"* (23). Some women had difficulty accessing information and understanding the self-management tasks, particularly when there were language barriers: *"It's a bit difficult because my English isn't very good…I have always been asking for an interpreter, but I have never got one"* (24). There were also concerns ranging from a lack of information (*"they didn't tell me what's the side effects for the baby"* (15)) to accusations of patronising healthcare professionals who *"talked down to [me]"* and *"didn't trust me"* (17).

Discussion

Principle Findings

This review found that women with diabetes predominantly undertake self-management during pregnancy for the health of their baby. Support networks are crucial in alleviating the burden that comes with managing a chronic condition, particularly a new one, such as gestational diabetes. When these networks are less effective, women may feel overwhelmed by the self-management, which is compounded by a lack of knowledge and understanding.

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One of the most notable findings from this review was the lack of research dedicated to managing chronic conditions in pregnancy. Diabetes was the only condition represented in the included literature, but with only 16 papers eligible for inclusion, the qualitative evidence base around managing gestational or chronic diabetes in pregnancy is still thin. While the findings of this review are framed by the self-management challenges of diabetes, a number of generic activities are applicable across other chronic conditions. Self-management of diabetes includes self-testing (of blood glucose levels) and diet and exercise management, all of which are applicable more widely in pregnancy. However, it is clear there is a lack of specific research on other chronic conditions in pregnancy, such as hypertension. It is also worth noting that in the general population chronic conditions are self-managed due to the absence of a 'cure', but this is not strictly the case in pregnancy where conditions are temporarily bounded by the pregnancy; gestational diabetes and hypertension (or pre-eclampsia) can be resolved with delivery, and if they are not resolved, they are re-categorised as type 2 diabetes or hypertension (25) – thus, they are no longer conditions brought about by pregnancy.

The results of this review have demonstrated that the motivating factors for pregnant women to self-manage are different to those motivating the general population; none more so than the desire to improve outcomes for the baby. As far as the authors are aware this paper is the first to use the burden of treatment theory as a theoretical lens to explore experiences of chronic disease management in pregnancy. Although this theory does not fully account for the specific challenges of pregnancy, as demonstrated by this review, it still broadly accounts for the actors and work involved in self-management in this context. Using this theory allowed for a deeper understanding of the burden of self-management and the work involved.

Interpretation

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Acting for the Baby

May et al argue that the work of self-management in the general population is often done in the context of "*holding together something larger and more complex*" (14), alluding to everyday life. While this applies in pregnancy, the work of self-management is also done in the context of maintaining the health of the unborn baby. Previous literature has touched on pregnant people acting "*not for the self but also for the unborn "other"*' (26) and the burden of pregnant women having to be accountable for the health of their baby alongside their own health cannot be underestimated. Current self-management models are not equipped to consider that particular burden of treatment.

Support Networks

Current self-management models emphasise that "*self-management is manifested as both an individual and family construct*" (27) and underscore the importance and necessity of social networks. The importance of social networks is also realised in the Burden of Treatment theory. May asserts that "*knowledge and beliefs about health and healthcare are often shared [within social networks]…decisions about what to do, and how to access services, are often distributed amongst multiple participants in a social process*" (14). In pregnancy, the sharing of lay knowledge and expertise is commonplace. Hinton et al demonstrated the nature of information sharing among pregnant people, often now in the form of online communities, in those with pregnancy hypertension or raised blood pressure (28). Pregnant women with hypertension valued the advice and support of their peers, and in the online space peer advice tended to trump the advice of medical professionals (28). Support networks are a central part of any self-management model in the general population, often emphasising that "*self-management is manifested as both an individual and family construct*" (27). As both a barrier and a facilitator of self-management, it is

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evident family support is an important factor in continued adherence. While current self-management models do not accommodate pregnancy entirely, this review demonstrates that elements of existing models, such as social networks and the need for education around diagnosis, are applicable.

Strengths and Limitations

The research question evolved throughout the search as it became apparent that little data on self-management in pregnancy existed, and none outside of diabetes management. This review should act as a call for further work. Lived experiences are often over looked in self-management literature and by using qualitative literature we have been able to address some of these research gaps.

Limitations include that the coding framework is constrained by how the original researchers interpreted the interviews and their interactions with patients. Distinguishing between pre-existing chronic conditions and pregnancy acquired chronic conditions is important, but it was not a distinction that was reflected in the dataset; it is both a limitation of the included studies and this review. While the majority of the included papers were on gestational diabetes, there were two on pre-existing diabetes, and three with a mixed population of gestational and chronic diabetes. The burden of treatment theory does not perfectly map to pregnancy, however elements of this theory are helpful in understanding how and why patients interact with self-management. Many of the included papers specifically considered barriers to self-management, as opposed to facilitators, thus potentially creating a negative bias towards self-management.

Included studies took place across a variety of international settings, operating within different health systems, ranging from those dominated by private insurance-based systems to those with national healthcare systems aimed at universal healthcare coverage. Individualised medicine is intertwined with self-management; six of the included studies took place in the United States,

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where they operate within an insurance-based system that lends itself to an individualised approach to medicine. With the exception of Canada, New Zealand, and Singapore, none of the other countries included in this review have health systems based on almost near universal health care coverage. A patient's relationship with their healthcare system also affects their relationship with the acceptability of self-management and their efficacy undertaking it. As none of the included studies were conducted in the UK, it raises questions about how pregnancy self-management would operate within the NHS.

Conclusions

Clinical Implications

This review suggests there are several ways to improve adherence to self-management interventions in pregnancy, these interventions often involve self-monitoring and titrating medication. Health care professionals communicating the importance of self-management in terms that focus on the health of the baby, and educating women on the condition they have been diagnosed with, could have a positive impact on self-management adherence.

Research Implications

There is a need for research on the self-management of chronic conditions in pregnancy, particularly in chronic conditions other than diabetes. There are ongoing trials on self-management in pregnancy, predominantly in hypertension, but how self-management will fit into usual care remains to be seen. Remote forms of care are now more commonplace, in light of this, a better, qualitative understanding of the new burden of treatment self-management creates is needed.

Main Conclusions

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This review shows that the primary motivating factor for women self-managing is the health of their baby. Their support networks and their understanding of the condition contribute to whether they self-manage effectively. The burden of treatment shifting to women requires further research, as patient work increases, feelings of anxiety can also increase. In this dataset, it was clear that some women found this treatment shift to be overwhelming but their anxiety largely stemmed from a lack of knowledge and understanding of the condition with which they had been diagnosed. Evidently, education is a barrier to pregnant women effectively self-managing, as when these issues are addressed women are willing and able to self-manage.

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Contribution to Authorship

BJ, KT, LH, RM and NR contributed to creating the research question and parameters of the review, including the search strategy. BJ conducted the search with assistance from NR. BJ and JK reviewed the titles and abstracts of identified articles and carried out the full text assessments. Disagreements on reviewed articles were resolved by consensus or discussion with KT. Data extraction and complete analysis was conducted by BJ and reviewed by KT, LH and RM. The write up was led by BJ, with edits provided by KT, LH and RM.

Ethical approval

Not applicable.

Competing interests

The authors declare no conflict of interest.

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expressed in this publication are those of the authors and not necessarily those of the NHS, the NIHR, the Department of Health and Social Care, or the Primary Care Research Trust.

Patient Consent

Not required

Data Sharing Statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Public and Patient Involvement

Patients and the public were not involved in this systematic review. Since this is a qualitative systematic review of already published papers, and no participant recruitment took place, the involvement of patients and the public is not applicable.

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Tables

Table 1: Characteristics of Included Studies

Title	Year	Authors	Chronic Condition	Journal	Sample Size	Setting	Qualitative Method
Living with gestational diabetes in a rural community	2014	Abraham , K. Wilk, N.	Diabetes	The American Journal of Maternal and Child Nursing	10	USA	One on one interviews
Women's experiences of gestational diabetes self-management: A qualitative study	2013	Carolan, M.	Diabetes	Midwifery	15	Australia Diabetes clinic	One on one interviews, focus group
Women's experiences of factors that facilitate or inhibit gestational diabetes self-management	2012	Carolan, M. Gill, G. Steele, C.	Diabetes	BMC Pregnancy and Childbirth	15	Australia Diabetes clinic	One on one interviews, focus group
Experiences, perceptions and self-management of gestational diabetes in a group of overweight multiparous women	2014	Chavez-Courtois, M. Graham, C. Romeo-Perez, I. Sanchez-Miranda, G. Sanchez-Jimenez, B. Perichart-Perera, O.	Diabetes	Cien Saude Colet	5	Spain	One on one interviews
'Is gestational diabetes a severe illness?' exploring beliefs and self-care among women with gestational diabetes living in a rural area of the south east of China	2016	Ge, L. Wikby, K. Rask, M.	Diabetes	Australian Journal of Rural Health	17	China Obstetric clinics, wards	One on one interviews
The maternal experience of having diabetes in pregnancy	2011	Nolan, J. McCrone, S. Ilana, R. Chertok, A.	Diabetes	Journal of the American Academy of Nurse Practitioners	8	USA	Focus groups
Self-monitoring lifestyle behaviour in overweight and obese pregnant women: qualitative findings	2018	Sheih, C. Draucker, C.B	Overweight /obese	Journal of Clinical Nursing	13	USA Clinics (community health centre and health network)	One on one interviews
Socio-cultural aspects of self-management in gestational diabetes	2012	Wazqar, D. Evans, M.	Diabetes	Diabetes Nursing	12	Canada	Secondary analysis of one on one interviews
'I was tired of all the sticking and poking': Identifying barriers to diabetes self-care among low-income pregnant women	2015	Yee, L. McGuire, J. Taylor, S. Niznik, C. Simon, A.	Diabetes	Journal of Healthcare for the Poor and Underserved	10	USA Clinic	One on one interviews
Factors promoting diabetes self-care among low-income, minority pregnant women	2016	Yee, L. McGuire, J. Taylor, S. Niznik, C. Simon, A.	Diabetes	Journal of Perinatology	10	USA Diabetes clinic	One on one interviews
Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: a phenomenological research	2017	Youngwanichsetha, S. Phumdoung, S.	Diabetes	Journal of Clinical Nursing	30	Thailand Antenatal clinic, diabetes clinic, obstetric ward	One on one interviews
Self-management of gestational diabetes among Chinese migrants: A qualitative study	2018	Wah, Y. McGill, M. Wong, J. Ross, G. Harding, AJ. Krassa, I.	Diabetes	Women and Birth	18	Australia Antenatal clinic	One on one interviews
Ethnic Differences in Dietary Management of Gestational Diabetes Mellitus: A Mixed Methods Study Comparing Ethnic Chinese Immigrants and Australian Women	2019	Wan, C. Teede, H. Nankervis, A. Aroni, R.	Diabetes	Journal of the Academy of Nutrition and Dietetics	83	Australia Maternity services, tertiary hospitals	One on one interviews and diary entries
The experience of gestational diabetes for indigenous Māori women living in rural New Zealand: qualitative research informing the development of decolonising interventions	2018	Reid, J. Anderson, A. Cormack, D. Reid, P. Harwood, M.	Diabetes	BMC Pregnancy and Childbirth	10	New Zealand Primary care clinic	One on one interviews

"Diabetes Just Tends to Take Over Everything": Experiences of Support and Barriers to Diabetes Management for Pregnancy in Women With Type 1 Diabetes	2019	Singh, H. Ingersoll, K. Gonder-Frederick, L. Ritterband, L.	Diabetes	Diabetes Spectrum	15	USA Diabetes clinic	One on one interviews
Barriers to Gestational Diabetes Management and Preferred Interventions for Women With Gestational Diabetes in Singapore: Mixed Methods Study	2020	Hewage, S. Audimulam, J. Sullivan, E. Chi, C. Yew, T. W. Yoong, J.	Diabetes	JMIR Formative Research	15	Singapore Gestational Diabetes clinic	One on one interviews, in a mixed methods study also including a survey

Table 2: Quote Table: Barriers and Motivations for Self-management

Barriers to self-management		Motivations for self-management	
Lack of knowledge and understanding	"Actually, I didn't know anything about it...So, I...thought that if you were diabetic then you kind of get it. But then I didn't know something you can just develop during pregnancy as well. So, it was quite new to me." (29)	**Desire for healthy baby**	"I have a responsibility to care for my child…and to care for this baby inside of me…For me, I wasn't going to let anything get in the way…" (30)
Feeling anxious and overwhelmed	"With the fourth pregnancy, I had sort of lost control. I was living in the garage with my three children [and husband]. I was huge and six-months [along in my pregnancy] when I was diagnosed with depression….I didn't want to do anything. I'd just wake up [and] went through the motions." (31)	**Supportive environment: family**	"If it's not because of my husband, I couldn't have made it this far, like he would remind me, sometimes when I want to eat something he would just remind me not to, and like when we go for shopping he would buy the healthy stuff." (24)
Lack of support: family	"I need to cook for my husband and I can't make him eat the same thing…I need to follow him because he needs energy for his work…and I can't avoid using sauce, you know how on the menu it says you need to avoid sauces and stuff, I can't do that, my husband wouldn't want to eat according to that." (24)	**Supportive environment: HCP**	"The diabetes educator is really friendly...she explained things very like, in a very good way Yes, yes and – like, she did a demo in front of me, how to inject yourself. It was really scary first time. They told me everything." (19) "The first pregnancy I had a lady (educator) that was really excellent, very understanding, very approachable. She would listen to my concerns and we had a conversation rather than just a one-way flow of information. And so I had a very positive experience with her." (15)
Lack of support: HCP	"You are told that you have this and nobody spends time with you, it's kind of frustrating because you want to cry. You think that it's something you did wrong" (17) "According to my experience, what they (health professionals) discussed are all about the dietary habit of women here (in Australia), so if the suggestions could be more relevant to Chinese lifestyle, a lot of things that have been taught currently are not meaningful." (20)		

Figures

Figure 1: PRISMA Flow Diagram

Legend: PRISMA Flow Diagram of study search strategy

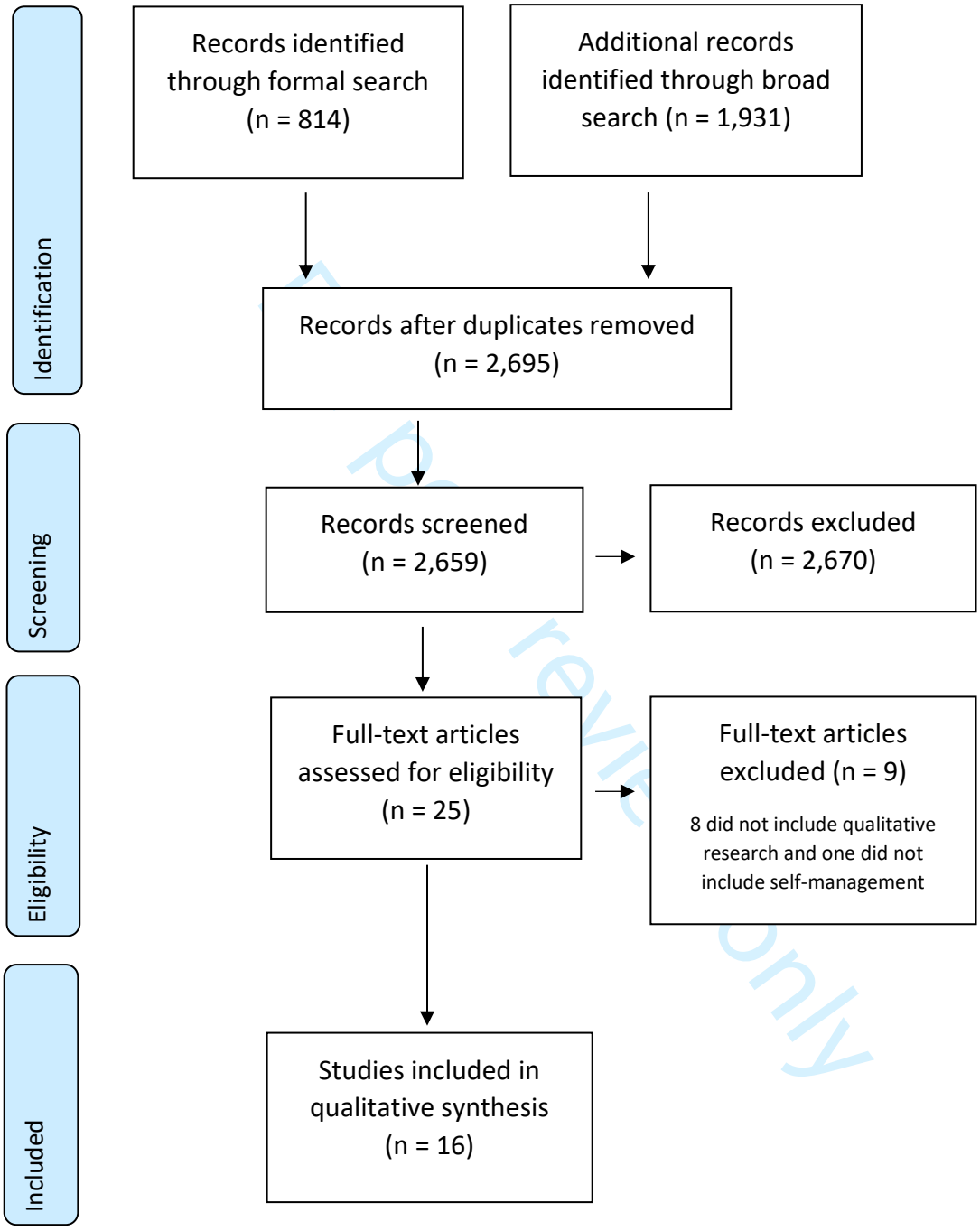
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Identification
Screening
Eligibility
Included
Records identified through formal search (n = 814)
Additional records identified through broad search (n = 1,931)
Records after duplicates removed (n = 2,695)
Records screened (n = 2,659)
Records excluded (n = 2,670)
Full-text articles assessed for eligibility (n = 25)
Full-text articles excluded (n = 9)
8 did not include qualitative research and one did not include self-management
Studies included in qualitative synthesis (n = 16)

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International prospective register of systematic reviews

National Institute for Health Research

University of York
Centre for Reviews and Dissemination

??? For those soon to register with PROSPERO, records may not be submitted between Thursday 1st April (5 pm) and Tuesday 6th (8 am) whilst staff are away and the university is closed for the Easter period. During this time you may still work on your record and save it for submission after Easter.

Systematic review

1. * Review title.

Give the title of the review in English

What are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?

2. Original language title.

For reviews in languages other than English, give the title in the original language. This will be displayed with the English language title.

3. * Anticipated or actual start date.

Give the date the systematic review started or is expected to start.

01/04/2019

4. * Anticipated completion date.

Give the date by which the review is expected to be completed.

31/12/2019

5. * Stage of review at time of this submission.

Tick the boxes to show which review tasks have been started and which have been completed. Update this field each time any amendments are made to a published record.

Reviews that have started data extraction (at the time of initial submission) are not eligible for inclusion in PROSPERO. If there is later evidence that incorrect status and/or completion date has been supplied, the published PROSPERO record will be marked as retracted.

This field uses answers to initial screening questions. It cannot be edited until after registration.

The review has not yet started: No

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NHS National Institute for Health Research

Review stage	Started	Completed
Preliminary searches	Yes	No
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	Yes	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Provide any other relevant information about the stage of the review here.

6. * Named contact.

The named contact is the guarantor for the accuracy of the information in the register record. This may be any member of the review team.

Beth Jakubowski

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

Ms Jakubowski

7. * Named contact email.

Give the electronic email address of the named contact.

bethany.jakubowski@phc.ox.ac.uk

8. Named contact address

Give the full institutional/organisational postal address for the named contact.

9. Named contact phone number.

Give the telephone number for the named contact, including international dialling code.

01865 617961

10. * Organisational affiliation of the review.

Full title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

University of Oxford

Organisation web address:

https://www.phc.ox.ac.uk/

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National Institute for Health Research

11. * Review team members and their organisational affiliations.

Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. **NOTE: email and country now MUST be entered for each person, unless you are amending a published record.**

Ms Bethany Jakubowski. University of Oxford
Ms Jaspreet Khaira. University of Oxford
Professor Richard McManus. University of Oxford
Dr Katherine Tucker. University of Oxford
Dr Lisa Hinton. University of Cambridge

12. * Funding sources/sponsors.

Details of the individuals, organizations, groups, companies or other legal entities who have funded or sponsored the review.

CLAHRC Oxford

Grant number(s)

State the funder, grant or award number and the date of award

13. * Conflicts of interest.

List actual or perceived conflicts of interest (financial or academic).

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country must be completed for each person, unless you are amending a published record.**

15. * Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

What are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

The search terms used to find articles in PubMed

- Pregnancy MeSH
- Pregnant women MeSH
- Pregnan* OR gestational OR maternal
- Decision making MeSH
- Self-care MeSH
- Shared decision making OR self manag* OR self-monitor

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The search terms used for CINAHL

- Pregnancy OR Expectant Mothers

- Pregnan* OR gestational OR maternal

- Self-care

- Shared decision making OR self-manag* OR self-monitor

There was also a broad secondary search using the search term 'pregnancy self-management' in PubMed

17. URL to search strategy.

Upload a file with your search strategy, or an example of a search strategy for a specific database, (including the keywords) in pdf or word format. In doing so you are consenting to the file being made publicly accessible. Or provide a URL or link to the strategy. Do NOT provide links to your search **results**.

https://www.crd.york.ac.uk/PROSPEROFILES/136681_STRATEGY_20200916.pdf

Alternatively, upload your search strategy to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

Do not make this file publicly available until the review is complete

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

This review is on the self-management of chronic conditions in pregnancy i.e. hypertension, diabetes, kidney disease. The scope of the review is to look at the feasibility and acceptability of women self-managing, and how they regard this new burden of treatment.

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

The population being studied is pregnant women with a chronic condition, there are no further restrictions on the study population.

Inclusion Criteria

1. Chronic condition: diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease, others to be reviewed as they come up

2. Pregnancy

3. Self-management: self-monitoring, self-weighing, self-care, others to be reviewed as they come up

4. Qualitative.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The

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preferred format includes details of both inclusion and exclusion criteria.

This review is looking at the self-management of chronic conditions in pregnancy. Self-management can be defined as self-monitoring (i.e. of blood glucose or blood pressure), or as action on the basis of self-monitoring (i.e. diet management, self-care, weight management).

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

Usual care if appropriate.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Qualitative studies are eligible for inclusion, including secondary analysis of previous qualitative research.

Inclusion Criteria

1. Chronic condition: diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease, others to be reviewed as they come up

2. Pregnancy

3. Self-management: self-monitoring, self-weighing, self-care

4. Qualitative

23. Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

Chronic conditions in pregnancy are under researched, so the aim of this review is to examine the current qualitative evidence of the behaviours and attitudes towards the self-management of chronic conditions in pregnancy. It is important to determine what, if any, self-management currently takes place and whether women find it acceptable and feasible to self-manage. Therefore the inclusion criteria for this review is for the participant to have a chronic condition (i.e. hypertension, diabetes) during pregnancy (this can be defined as gestational or chronic), and for some form of self-management to be taking place. This self-management can either be self-monitoring (i.e. of blood glucose or blood pressure), or action on the basis of self-monitoring (i.e. diet management, self-care, weight management).

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

The study aims to explore whether self-management of chronic conditions in pregnancy is acceptable and

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feasible to pregnant women. It also aims to explore current practice and the current evidence base of chronic disease management in pregnancy.

Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

None.

Measures of effect

Please specify the effect measure(s) for you additional outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

All of the studies identified in the search were screened independently by two reviewers (title and abstract) who then came together to agree on which articles should get a full text review. Both reviewers then independently completed the full text review and met again to agree on which articles should be included in the final review. Discrepancies were resolved by reviewing the definition of self-management and chronic condition as these were often the causes of uncertainty. The first author will then use a data extraction spreadsheet to list the key themes and findings from the papers. 10% of the included articles will be screened by another member of the study team, measuring the 10% of articles against the inclusion criteria and the COREQ quality assessment guidelines. The included studies will then be put into NVivo 12 and a thematic analysis, using the guidelines produced by Braun and Clarke (2006), will be carried out by the first author. A theoretical lens will be used to reflect the themes that emerge during data analysis.

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

Two researchers carried out the screening of articles, and met after the title and abstract screening to feed back results and decide which articles required a full text review. They met again following this screening to decide which articles should be included in the final review. To determine the quality of the studies, the COREQ guidelines will be used, which is a standard set of guidelines to determine the quality of qualitative work. The quality of the studies will be acknowledged in the write up, in order to be transparent about the

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quality of the work in this field to date.

28. * Strategy for data synthesis.

Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta-analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.

Using NVivo 12, an initial thematic analysis will be conducted by the first author to create a coding framework of themes and sub themes. A theoretical lens will be used to develop the coding framework further. Discrepancies will be discussed with the wider study team and feedback will be incorporated into the framework.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

Not currently planned

30. * Type and method of review.

Select the type of review, review method and health area from the lists below.

Type of review

Cost effectiveness
No

Diagnostic
No

Epidemiologic
No

Individual patient data (IPD) meta-analysis
No

Intervention
No

Meta-analysis
No

Methodology
No

Narrative synthesis
No

Network meta-analysis
No

Pre-clinical
No

Prevention
No

Prognostic

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No

Prospective meta-analysis (PMA)
No

Review of reviews
No

Service delivery
No

Synthesis of qualitative studies
Yes

Systematic review
Yes

Other
No

Health area of the review

Alcohol/substance misuse/abuse
No

Blood and immune system
No

Cancer
No

Cardiovascular
No

Care of the elderly
No

Child health
No

Complementary therapies
No

COVID-19
No

Crime and justice
No

Dental
No

Digestive system
No

Ear, nose and throat
No

Education
No

Endocrine and metabolic disorders
No

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NHS National Institute for Health Research

Eye disorders
No

General interest
No

Genetics
No

Health inequalities/health equity
No

Infections and infestations
No

International development
No

Mental health and behavioural conditions
No

Musculoskeletal
No

Neurological
No

Nursing
No

Obstetrics and gynaecology
No

Oral health
No

Palliative care
No

Perioperative care
No

Physiotherapy
No

Pregnancy and childbirth
Yes

Public health (including social determinants of health)
No

Rehabilitation
No

Respiratory disorders
No

Service delivery
No

Skin disorders
No

Social care

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No

Surgery
No

Tropical Medicine
No

Urological
No

Wounds, injuries and accidents
No

Violence and abuse
No

31. Language.

Select each language individually to add it to the list below, use the bin icon to remove any added in error.
English

There is not an English language summary

32. * Country.

Select the country in which the review is being carried out. For multi-national collaborations select all the countries involved.

England

33. Other registration details.

Name any other organisation where the systematic review title or protocol is registered (e.g. Campbell, or The Joanna Briggs Institute) together with any unique identification number assigned by them. If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34. Reference and/or URL for published protocol.

If the protocol for this review is published provide details (authors, title and journal details, preferably in Vancouver format)

Add web link to the published protocol.

Or, upload your published protocol here in pdf format. Note that the upload will be publicly accessible.

No I do not make this file publicly available until the review is complete

Please note that the information required in the PROSPERO registration form must be completed in full even if access to a protocol is given.

35. Dissemination plans.

Do you intend to publish the review on completion?

Yes

Give brief details of plans for communicating review findings.?

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We will aim to publish the review in a high impact scientific journal.

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords help PROSPERO users find your review (keywords do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

Pregnancy; maternal health; chronic conditions; self-management

37. Details of any existing review of the same topic by the same authors.

If you are registering an update of an existing review give details of the earlier versions and include a full bibliographic reference, if available.

38. * Current review status.

Update review status when the review is completed and when it is published.New registrations must be ongoing so this field is not editable for initial submission.
Please provide anticipated publication date

Review_Ongoing

39. Any additional information.

Provide any other information relevant to the registration of this review.

40. Details of final report/publication(s) or preprints if available.

Leave empty until publication details are available OR you have a link to a preprint (NOTE: this field is not editable for initial submission). List authors, title and journal details preferably in Vancouver format.

Give the link to the published review or preprint.

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Appendix 1: Search Strategy

Two databases were searched, CINAHL and Pub Med. The search terms used to find articles in Pub Med (n=107) were MeSH terms 'pregnancy,' 'pregnant women', 'decision making', 'self-care', and 'pregnan* OR gestational OR maternal', and 'shared decision making OR self manag* OR self-monitor'. There was also a broad secondary search in Pub Med using the search term 'pregnancy self-management' which identified a further 1,931 articles for review. The search terms used for CINAHL (n=707) were 'pregnancy OR expectant mothers', 'pregnan* OR gestational OR maternal', 'self-care', and 'shared decision making OR self-manag* OR self-monitor'. This identified a total of 2,745 articles, which when controlled for duplicates left 2,695 for review. 25 articles were identified for full text review, of which 16 were selected for the systematic review.

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Quality Appraisal Checklist	Personal Characteristics				Relationship with Participants			Theoretical Framework	Participant Selection				Setting			Data collection						
Title	Interviewer/Facilitator	Credentials	Occupation	Experience and training	Relationship established	Participant knowledge of the interviewer/interview	Interviewer characteristics	Methodological orientation and theory	Sampling	Method of approach	Sample size	Non-participation	Setting of data selection	Presence of non-participants	Description of sample	Interview guide	Repeat interviews	Audio/visual recording	Field notes	Duration	Data saturation	Transcripts returned
Self-monitoring lifestyle behaviour in overweight and obese pregnant women: qualitative findings	First author	Author 1: DNSc, MPH, RNC-OB, FAAN. Author 2: PhD, RN, FAAN	Author 1: Associate Professor. Author 2: Researcher	First author: Expert in prenatal health promotion and content analysis. Second author: PhD. Doctoral student carrying out analysis: undergraduate degree in exercise science, experience in a home visitation programme for low-income pregnant women	First author called or e-mailed each woman 3 - 7 days before a scheduled interview	Not mentioned - assumed through recruitment	No bias reported	Self-monitoring theory, behavioural and lifestyle change theories	Purposive	Interview explained to trial participants and informed consent sought	13	22 women enrolled in the intervention, 9 refused interview	10 interviewed in clinic private rooms, 3 over phone	No	Mean age 29.5 years, majority obese, majority were African American (77%), unmarried (69%), previous pregnancy (77%), third trimester (69%)	Yes	No	Audio	No	30 minutes	Not discussed	No
"I was tired of all the sticking and poking": Identifying Barriers to Diabetes Self-Care Among Low-Income Pregnant Women	Author 1 and 2	Author 1: MD, MPH, Author 2: MD, MPH, Author 3: MPH, Author 4: APN, CDE, Author 5: MD, MPH	Author 1: Fellow in Maternal-Fetal Medicine. Author 2: Resident in OBGYN at Northwestern. Author 3: Senior Research Coordinator at Northwestern. Author 4: Advance Practice Nurse in Maternal-Fetal Medicine. Author 5: Professor of OBGYN at Northwestern	Extensive expertise in maternal-fetal medicine and research	Not mentioned	Were identified because they had GDM so knew why the research was being conducted	No bias reported	Cognitive load theory	Purposive	At a clinic serving public-assisted low-income women. Informed consent sought.	10	46 women identified, 22 ineligible, 12 declined participation	Conducted in private	No	Mean age 28.5, African American (4), Latina (4), Native American (1), Asian (1). Most educated to high school degree level, most multiparous, all diabetic	Yes	Three interviews per person	Audio	No	30-60 minutes	Reached at 29 interviews	No
The maternal experience of having diabetes in pregnancy	Not reported by coder/analyser was present for all interviews and focus groups	Author 1: MD, Author 2: PhD, RN, PMHCNS-BC, Author 3: PhD, RN, IBCLC	Not reported	Extensive expertise judging by credentials	Not mentioned	Recruitment flyers provided some knowledge of subject matter	No bias reported	Phenomenology	Convenience	Flyers distributed in healthcare provider offices	8	Not reported	Over the phone or in person	Yes, additional investigator to take field notes	Average age 35, mostly highly educated, no demographics reported	Yes	No - these were focus groups	Audio	Yes	Not reported	Not discussed	Some were to check participants views were represented correctly
Women's experiences of gestational diabetes self-management: a qualitative study	Not reported	Author 1: PhD	Associate Professor Midwifery	Not reported but has a PhD in a related field	Approached by researcher prior to interview to confirm interest	Recruited from diabetes clinic fitting criteria	No bias reported	Phenomenology	Purposive (blinded for review)	Women invited to participate if fitted criteria, approached by researcher if Diabetes Coordinator confirmed their interest	15	Not reported	10 phone interviews, one in interviewee's home, one focus group at the hospital clinic	No	Variety of ethnic backgrounds, 23 - 40, majority high school educated, majority primiparous	Not reported	No	Audio	No	Not reported	Not discussed	No
Factors promoting diabetes self-care among low-income, minority pregnant women	Author 1 and 2	Author 1: MD, MPH, Author 2: MD, MPH, Author 3: MPH, Author 4: APN, CDE, Author 5: MD, MPH	Author 1: Fellow in Maternal-Fetal Medicine. Author 2: Resident in OBGYN at Northwestern. Author 3: Senior Research Coordinator at Northwestern. Author 4: Advance Practice Nurse in Maternal-Fetal Medicine. Author 5: Professor of OBGYN at Northwestern	Members of the research team reported to have prior qualitative research experience and extensive interviewing training	Not mentioned	Were identified because they had GDM and at their first clinic visit so knew why the research was being conducted	No bias reported	Cognitive load theory	Purposive	Recruited at their first clinic visit for diabetes care	10	46 women identified, 22 ineligible, 12 declined participation	Conducted in private	No	Mean age 28.5, African American (4), Latina (4), Native American (1), Asian (1). Most educated to high school degree level, most multiparous, all diabetic	Yes, provided in article	Three interviews per person	Audio	No	30-60 minutes	Reached at 29 interviews	No
Socio-cultural aspects of self-management in gestational diabetes	Author 1	Author 1: PhD student. Author 2: Assistant Professor	Author 1: PhD student. Author 2: Assistant Professor Nursing	PhD student	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Phenomenology	Purposive	Women being treated for GD were recruited	12	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Aged between 23 - 38, most worked outside the home, most were married, mix of first pregnancy and second pregnancies	Yes	Two interviews conducted per person	Audio	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis
Women's experiences of factors that facilitate or inhibit gestational diabetes self-management	Author 1	Not reported	Not reported	Not reported	Access to women facilitated by diabetes co-ordinator, no relationship pre-established with researcher	Recruited through diabetes clinic and by diabetes co-ordinator so would have been told what the study was about	No bias reported	Interpretative phenomenology	Purposive	Diabetes coordinator approached women, participants were recruited after they attended GDM education and had at least 3 weeks self management of condition	15	30 women met inclusion criteria, 20 agreed to participate, 15 available on day of interview	Focus group in private room in clinic, phone interviews, face to face interview at one woman's home	No	Age range 23 -40, education level lower than population average, mix of ethnic backgrounds, majority of women primiparous	Not reported	No	Audio	No	Not reported	Not mentioned	No
Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: a phenomenological research	Not reported	Author 1: PhD, RN, Assistant Professor of Nursing. Author 2: PhD, RN, Associate Professor of Nursing	Author 1: Assistant Professor. Author 2: Associate Professor	Not reported	Prolonged engagement was carried out by researchers to build a rapport with participants before interviews	Participants approached by staff nurses so knew about what the study was about prior to taking part	No bias reported	Phenomenology	Purposive	Approached at antenatal care clinics and informed consent obtained	30	Not reported	Interviewed at the antenatal clinic, diabetes clinic or obstetric ward	No	The mean age of participants 32.5 years, 17 women were Buddhists, 13 women were Muslims. Majority multiparous and employed.	Yes	Two interviews conducted per person	Audio	No	30 - 45 minutes	Data were collected until saturation (after six months)	No
Living with gestational diabetes in a rural community	Not reported	Author 1: DNP, APRN-C. Author 2: DNS, WHNP-C	Author 1: Visiting Assistant Professor of Nursing. Author 2: Visiting Assistant Professor	Not reported	Established through recruitment	Had diagnosis of the disease which was being explored in interviews, and recruited via study posters	No bias reported	Phenomenology	Purposive and snowball	Information about study posted in obstetric and health care provider officers in New York State	10	Not reported	Interviewed in person or at phone	No	All women Caucasian between 25-49. All had history of GDM.	Yes	Follow up interviews conducted with four participants	Audio	No	15 - 30 minutes, second interviews 15 - 30 minutes	Checked by sharing core themes with participants	Transcripts of first interview provided to participants doing a second interview for verification
"Is gestational diabetes a severe illness?" Exploring beliefs and self-care behaviour among women with gestational diabetes living in a rural area of the south east of China	Author 1	Author 1: M, Msc. Author 2: RN, PhD. Author 3: RN, PhD	Author 1 a teacher of maternal care and bilingual	Not reported	Not mentioned	Interviewer did not know participants prior to study, was not part of the clinical care team.	No bias reported	No specific theoretical framework reported	Purposive	Women purposefully recruited from obstetric clinics and wards at a specific hospital.	17	Not reported	Interviewed in a room at the hospital	No	27.5 median age, six women high educational level, five middle educational level, six low educational level	Yes and provided in paper	No	Audio	No	40 - 60 minutes	Not mentioned	No
Experiences, perceptions and self management of gestational diabetes in a group of overweight multiparous women	Not reported	Not reported	Not reported	Interviews conducted by experts in anthropology and social psychology	Established through recruitment and screening process	Women were given a health questionnaire prior to recruitment	No bias reported	Phenomenology	Purposive	Women recruited through department of diabetes, given questionnaire and selected based on the questionnaire results	5	Not reported	Not reported	No	Reported in table	Yes	No	Audio	Yes	90 minutes	Not mentioned	No
Ethnic Differences in Dietary Management of Gestational Diabetes Mellitus: A Mixed Methods Study Comparing Ethnic Chinese Immigrants and Australian Women	Not reported	Author 1: MN, APD Author 2: PhD Author 3: MD Author 4: PhD	Not reported	Not reported	No	Were identified because they had GDM so knew why the research was being conducted	No bias reported	Symbolic interactionist framework and grounded theory	Purposive	Women recruited through GDM clinics	83	11 declined to participate and reasons were given	Not reported but some on the phone (others in person)	No	Reported in table	Yes	No	Audio	Yes	45 - 90 minutes	Not mentioned but large sample size so assumed there was saturation	No
Self-management of gestational diabetes among Chinese migrants: A qualitative study	Chief Investigator	Not reported	Faculties reported	CI was fluent in English, Mandarin and Cantonese	Participant's had interviewer's contact details	Interested participants were provided with written information that explained the background of the study and provided the contact details of the researchers. All participants were recruited after they had received diabetes education from a diabetes educator and had at least two weeks of self-management experience.	No bias reported	Thematic analysis	Purposive	Between August and October 2016, participants were purposively recruited from the antenatal clinic at the RPAH in the Inner West Region of Sydney, Australia. Bilingual flyers about the study were distributed at the clinic. Interested participants were provided with written information that explained the background of the study and provided the contact details of the researchers.	18	Not reported	The interviews were conducted by the chief investigator in either a staff meeting room or a waiting room at the hospital, according to the participant's preference.	Yes, participant's partner present in two interviews	Reported in table: The mean age of participants was 34 years (SD = 3.6; range 29–41). The majority of participants came from China, Hong Kong or Taiwan and held a university degree	Yes	No	Audio	No	Not reported	Not reported	No
The experience of gestational diabetes for indigenous Māori women living in rural New Zealand: qualitative research informing the development of decolonising interventions	Not reported	Not reported	Faculties reported	Not reported	Lead researcher spoke to participants ahead of the interviewer	The contact details of all consenting participants were then provided to the lead researcher, who contacted participants via phone or home visits (when no phone service was available). The lead investigator provided detailed information about the research, answered any questions from women and obtained written informed consent and confidentiality agreements (including accessing their diabetes-related health data from the clinic) for each participant.	No bias reported	Kaupapa Maoris Methodology	Purposive - a sub sample of a larger study	The ten participants advised that they either had a clinical history of GDM (n = 8) or had been exposed to diabetes in utero (n= 2). Eligible participants were recruited via the local primary care clinic between September and December of 2015.	10	Reported in the larger study (6 withdrew)	Interviews undertaken in participants' homes or the health centre, or a mutually agreed venue.	No	10 Māori women aged between 30 and 69 years. Outlined in full table, participants had diverse pregnancy experiences and current diabetes-related health conditions	Yes	No	Audio	No	Not reported	Not discussed	Participants were invited to review, edit or change their transcript prior to data analysis.
"Diabetes Just Tends to Take Over Everything": Experiences of Support and Barriers to Diabetes Management for Pregnancy in Women With Type 1 Diabetes	Not reported	Not reported	Not reported	The main interviewer had a background in health psychology and skills in motivational interviewing.	Potential participants could indicate interest by contacting the investigator. Study procedures and the consent process reviewed with them.	Potential participants were informed that the purpose of the study was to identify barriers to and supports for blood glucose management around pregnancy	No bias reported	Thematic analysis	Purposive	Participants were recruited using advertisement flyers that were sent for distribution to diabetes specific clinics around the United States. Adult U.S. women (≥18 years of age) with type 1 diabetes who were fluent in English and were either pregnant, planning pregnancy for the immediate future, or had experienced a previous pregnancy were invited to participate in a phone interview.	15	27 indicated interest	Decided to conduct the interviews over the phone unless participants were local to the study site and wished to come to the study office	No	All participants were married, with a mean age of 34 years and a mean duration of type 1 diabetes of 21 years. Ten participants used a continuous glucose monitoring (CGM) system, and 13 used an insulin pump to manage their diabetes. Ten women had previously been pregnant, three were pregnant at the time of the interview, and five were trying for or contemplating pregnancy in the immediate future (within 0–4 months from the time of their interview).	Yes	No	Audio	Yes - notes and reflections recorded by interviewer	Between 30 and 50 minutes	To achieve data saturation, 10–15 people were recruited, with the possibility of conducting additional interviews, if necessary	No
Barriers to Gestational Diabetes Management and Preferred Interventions for Women With Gestational Diabetes in Singapore: Mixed Methods Study	First and second author	Author 1, PhD, Author 2 MPH, Author 3 MPH, Author 4 MBBS, MD, Author 5, MBBS, Author 6, PhD	Faculties reported	Not reported	Established through recruitment	Recruited from diabetes clinic fitting criteria, participants had previously participated in a survey on the same subject.	No bias reported	Thematic analysis	Purposive	Potential participants for the survey (this is a mixed methods study and interview participants were recruited from survey participants) were approached in the waiting room of the GDM clinic. The study team refrained from approaching women who seemed tired or anxious. After getting informed consent, women were given a self-administered survey. For pregnant women who were willing to participate in prenatal interviews, sessions were scheduled based on their convenience. In the second phase, to increase survey completion, the women were given an incentive of $25 (US $17.5). In the first phase, 50 patients participated in the survey.	15	Of the 50 patients who completed the survey, 35 gave verbal consent to participate in the interviews and 15 participated in the interviews	Most interviews were conducted at the health care facility or at the participants' homes, in a private space conducive for the participant to share thoughts effectively.	No	The women who participated in interviews had a demographic profile similar to that of the survey questionnaire participants. Half of the 15 participants in the age group of 31 to 35 years were primiparous or had a family history of T2DM, and most of them were Chinese or Indian, employed full time, with a degree, and attending a private prenatal clinic.	Yes	No	Audio	No	Not reported	Reported at 15 interviews	No

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Data analysis					Reporting					Reasons
Number of data coders	Description of coding tree	Derivation of themes	Software	Participant checking	Quotations presented	Data and findings consistent	Clarity of major themes	Clarify of minor themes	Quality Assessment	
2	Yes	Themes derived from data	N/A	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	Medium	Small sample size, short interviews, however themes were clear and recruitment process was transparent
2	Yes	Themes derived from data	ATLAS.ti 6	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	High	Despite small sample size, multiple interviews allowed for a richer dataset.
1 but checked informally by members of research team	No	Themes derived from data and then refined based on existing evidence of topic	N/A	Yes	Some reported, no quote table	Yes	Clear	Clear	Medium	Small sample size, even though focus groups tend to be better with a smaller sample size, this was still quite low and does not allow for a rich data set and exploration of themes
1	No	Themes derived from data	N/A	No	Yes with a quote table	Yes	Clear	Clear	High	Good sample size, clear discussion of themes and recruitment
2	Yes	Themes derived from data	ATLAS.ti 6	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	High	Despite small sample size, multiple interviews allowed for a richer dataset.
1	Yes	Themes derived from data	Not mentioned	Not mentioned - secondary analysis	No quote table, in text quotes included	Yes	Clear	Clear	Medium	A good sample size for a secondary analysis, but more information on primary data collection needed for transparency
1	Yes	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	High	Good sample size and mixed qualitative methods, leading to an apparently rich dataset. Paper was clear.
1	Yes	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	High	Good sample size , rich dataset and exploration of themes. Clear paper.
Not reported	No	Themes derived from data	Not mentioned	Yes participants were informed of core themes for member checking and saturation verification	No quote table, in text quotes included	Yes	Clear	Clear	Medium	Small sample size, but the discussion of themes was nuanced and rich. The interviews were also reported as being quite short.
Not reported	No	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	Medium	Good sample size with rich, longer form interviews. Clear paper and recruitment process clearly explained.
Not reported	No	Themes derived from data	ATLAS.ti V.5	No	No quote table, in text quotes included	Yes	Clear	Clear	Low	Very small sample size, potentially too low to make any assumptions about experiences of self-management
At least 2	No	Themes derived from data	Nvivo 11	No	Yes and quote tables included	Yes	Clear	Clear	High	Very large sample size, very rich dataset, clear paper and all processes communicated well
2	No	Themes derived from data	Not mentioned	No	Yes, embedded in article	Yes	Clear	Clear	High	Good sample size, clear paper and clear recruitment and analysis processes described
The lead researcher independently coded the data, and discussed the codes and themes with two experiences kaupapa Māori researchers, who had access to the interview data	No	Themes derived from data	Not mentioned	Participants were invited to review, edit or change their transcript prior to data analysis.	Yes, embedded in article	Yes	Clear	Clear	Medium	Small sample size, but an exploratory study and acknowledged it was a sub set of women who are rarely interviewed about their experiences.
Each transcript reviewed and coded independently by two researchers	Yes	Themes derived from data	Not mentioned	No	Yes, embedded in article	Yes	Clear	Clear	High	Good sample size, clear description of analysis process and themes very clear and supplemented with quotes
1	Yes	Themes derived from data	Atlas Ti	No - but transcripts checked by an author for accuracy	Yes but quote tables not included	Yes	Clear	Clear	Medium	Based on the qualitative interviews alone the sample size is small. However, there was a rich discussion of themes and analysis was rigorous.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3, 4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5, 6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5, 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7, 8, 10, 25, 26
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7, 8, 10, 25, 26
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7, 8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	N/A
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9, 10, 11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10 - 14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15, 16, 17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17, 18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18, 19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19, 20

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BMJ Open

Is self-management a burden? What are the experiences of women self-managing chronic conditions during pregnancy? A systematic review.

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Is self-management a burden? What are the experiences of women self-managing chronic conditions during pregnancy? A systematic review.

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Abstract

Objective

This systematic review examines the qualitative literature on women's experiences of self-managing chronic conditions in pregnancy.

Design

Systematic review of qualitative literature. Searches were performed in PubMed and CINAHL from inception to February 2021. Critical interpretive synthesis informed the coding framework and the analysis of the data. The Burden of Treatment theory emerged during the initial analysis as having the most synergy with the included literature, themes were refined to consider key concepts from this theory.

Participants

Pregnant women who are self-managing a chronic condition.

Results

A total of 2,695 articles were screened and 25 were reviewed in detail. All 16 included studies concerned diabetes self-management in pregnancy. Common themes coalesced around motivations for, and barriers to, self-management. Women self-managed primarily for the health of their baby. Barriers identified were anxiety, lack of understanding, and a lack of support from families and healthcare professionals.

Conclusions

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Pregnant women have different motivating factors for self-management than the general population and further research on a range of self-management of chronic conditions in pregnancy is needed.

Prospero Registration Number

CRD42019136681

Keywords: Systematic reviews, pregnancy, diabetes in pregnancy, chronic conditions, qualitative research, antenatal care, self-management, self-care, self-efficacy

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Article Summary

Short Title: Is self-management a burden?

Strengths and limitations of this study

- Pregnant women's experiences of self-managing during pregnancy is under researched compared to the general population. Lived experiences are often over looked in self-management literature and by using qualitative literature we have been able to address some of these research gaps.
- Limitations include that the coding framework is constrained by how the original researchers interpreted the interviews and their interactions with patients.
- Distinguishing between pre-existing chronic conditions and pregnancy acquired chronic conditions is important, but it was not a distinction that was reflected in the dataset; it is both a limitation of the included studies and this review.
- The burden of treatment theory does not perfectly map to pregnancy, however elements of this theory are helpful in understanding how and why patients interact with self-management.

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Introduction

Chronic diseases that manifest during pregnancy, or exist pre-pregnancy, can have a lasting impact on the health of the mother (1). Numerous conditions can affect women during pregnancy, some confined to that time (for example pre-eclampsia or gestational diabetes) and others that are longer lasting (chronic hypertension and diabetes). These conditions may lead to harm for both the mother and the baby; hypertensive disorders, for example, are one of the leading causes of maternal mortality globally (2).

Self-management

Self-management has been shown to support and improve the management of chronic conditions in the general population and there is a growing body of research to suggest self-management of chronic conditions in the pregnant population is also effective (3) (4) (5). Self-management interventions in pregnancy tend to focus on self-monitoring (blood pressure, blood glucose etc.) and in some cases titrating medication (insulin). Self-management following a hypertensive pregnancy has been explored post-partum, the evidence to date suggests not only is it feasible, but it may result in better diastolic blood pressure control at six months post-partum (4). In the antenatal period, the OPTIMUM-BP pilot trial suggested that self-monitoring of blood pressure was both feasible and acceptable to women with chronic hypertension (5). Self-management of diabetes in pregnancy has also been explored. In pregnant women with type 1 and type 2 diabetes, self-monitoring of blood glucose is common (6) and self-management programmes in women with gestational diabetes are also becoming more commonplace (3) (7). In the general population, self-management has been explored in numerous chronic conditions; in particular hypertension and diabetes, where there is a large body of research around understanding the patient experience (8) (9) (10). Much of this research has demonstrated patient's confidence and ability to self-manage

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(sometimes referred to as self-efficacy) (8) and suggests that self-management strategies improve health outcomes for those with chronic conditions (11). These conditions also affect women during pregnancy, but thus far this population are under-represented in this field of research. Moreover, research on self-management in the general population may not be wholly applicable to the pregnant population. Pregnancy presents an additional set of motivations for self-management, not accounted for in current self-management frameworks which often assume long-term self-management, whereas with pregnancy the period of management is often temporarily bounded even if it continues post-partum and considerations are for the mother and baby dyad.

Methods

This review was influenced by the critical interpretive synthesis (CIS) approach (12). A CIS approach leaves more space for interpretation than a conventional systematic review and in theory using this approach allowed us more space to consider lived experiences across a range of chronic conditions. This approach also allowed us to iteratively develop the research question and the inclusion criteria as the search progressed and the lack of research on the self-management of chronic conditions in pregnancy became apparent. In light of a lack of data, it was necessary to limit the scope of the review to the following research question: “what are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?”

We registered this review in PROSPERO (CRD42019136681, see Appendix 1, Supplementary File). Searches were performed in PubMed and CINAHL [EBSCOHost] (1982-present) from inception to February 2021 (see Appendix 2, Supplementary File).

Potentially eligible studies were screened by two independent reviewers against the following inclusion criteria:

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1. Chronic condition: including but not exclusively diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease.
2. Pregnancy.
3. Self-management: including but not exclusively self-monitoring, self-weighing, self-care.
4. Qualitative methods.

This review used a broad definition of self-management, "*the ability of an individual, in conjunction with family, community and health care professionals, to manage symptoms, treatments and lifestyle changes*" (13). We expanded the search criteria retrospectively to include risk factors for chronic conditions, in order to capture as much data as possible. The authors felt that qualitative literature was best placed to explore the lived experiences of people self-managing during pregnancy. Papers were excluded if the reviewers concluded that commonly recognised qualitative methods were not used, i.e. interviews, focus groups, or ethnographic observations.

Two reviewers (BJ & JK) independently reviewed the titles and abstracts of identified articles, a full text assessment of the relevant papers by both reviewers followed. The reference lists of included studies were screened but found no further articles to include. Disagreements on articles were resolved by consensus or discussion with a third reviewer (KT). Data extraction was conducted by BJ and checked by the study team.

Data Analysis

CIS also informed the analysis of the data (12). Analysis began with a detailed examination of the included papers, identifying themes and developing an understanding of the material. Dixon-Woods et al emphasise what distinguishes a systematic review using a CIS approach is *"the aim of being critical*" and questioning "*taken-for-granted-assumptions*" (12) meaning that the coding framework for this review was developed to critique the idea that general population self-

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management research was applicable to pregnancy. During the initial analysis, elements of the Burden of Treatment theory, developed for chronic condition management in the general population, were identified in the included literature. The coding framework was revised considering this theory's key concepts, such as the effect of patient work (e.g. self-monitoring, self-management, lifestyle changes) on self-management efficacy and the role of support networks, allowing for a more nuanced interpretation of the woman's experience of self-management. Data extraction was conducted by BJ using NVivo 12 Pro (QSR International Pty Ltd, released 2018), and a coding framework was developed by BJ and agreed upon by the study team (LH, KT, RM).

The Burden of Treatment

May et al have developed a theory explaining the relationship between "sick people" and their healthcare providers in the context of chronic disease. In the absence of a possible cure, patients must engage in 'routine work' to manage the disease (14). May et al explain that patient work in the context of the Burden of Treatment theory includes symptom monitoring, treatment management, and lifestyle changes while "*maintaining* [these tasks] *alongside the demands of other aspects of everyday life*" (14). Transferring the burden of treatment to the patient prompts a change in the patient-clinician relationship. Within pregnancy this could theoretically provide women with more agency, as well as more responsibility.

The first interpretation of the data, reviewing the themes that already existed in the included papers, revealed the synergy with the Burden of Treatment theory; therefore, the coding framework was iteratively mapped to two broad themes reflecting elements of the burden of treatment. These themes were grouped around motivations for self-management and barriers to self-management. As the coding progressed, more elements of the theory were incorporated into the analysis. One element

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of the theory is "*the structure and performance of patient work*" a process that May et al call "*sense making*" where "*people and members of their social networks are expected to identify, understand and explain the diverse tasks that make up their work*" (14). This idea was reflected in the themes concerning support from health care professionals and family, the impact of a lack of support, and women's anxiety and understanding of their diagnosis.

Results

Searches in PubMed and CINAHL identified 2,745 articles, which when controlled for duplicates left 2,695 for screening (see Appendix 2 and Figure 1 for further details, Supplementary File). The majority of articles were excluded as they did not include qualitative research or self-management, or the study population was not pregnant women. 25 articles were identified by the two reviewers for full text review, of which 16 fulfilled the inclusion criteria (see Table 1). Data quality checks using the COREQ guidelines were carried out (see Appendix 3, Supplementary File). Fifteen of the included studies were of medium to high quality. One was considered low quality because of the small sample size.

Included studies were published between 2011 and 2020 and all focused on diabetes in pregnancy. Six were conducted in the United States, four in Australia, and the remaining studies were carried out in Canada, Spain, Thailand, China, Singapore and New Zealand. The studies recruited from a range of settings: gestational diabetes clinics, antenatal clinics supporting low-income women, health care provider's offices, day care centres, and hospital based obstetric clinics. Fifteen of the studies used one-to-one interviews and one used focus groups for primary data collection. Two studies supplemented the interviews with focus groups, one supplemented with additional diary entries. One study was a mixed methods study, including one-to-one interviews alongside survey data. The study population for all included studies was pregnant women, although one study had a

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mixed population of women who were currently pregnant, women planning on getting pregnant, and women who had previously been pregnant. Only papers concerning diabetes self-management in pregnancy fulfilled the inclusion criteria, leading to the results focusing on the challenges of self-management in this disease.

The two broad themes that emerged from the data were motivations for and, conversely, barriers to self-management during pregnancy (see Table 2). Under the former, women express that having a healthy baby, supportive families, and supportive health care professionals were their primary motivating factors. Barriers to self-management included a lack of knowledge and understanding of the condition lead to feelings of anxiety, as well as a lack of support from families and healthcare professionals. The following sections will address each sub-theme in detail.

Motivations for self-management

Desire for a healthy baby

The most significant motivating factor for self-management was the desire for a healthy baby. Pregnancy comes with specific concerns and anxieties, women felt anxious over the burden of self-management and the health of their baby. It was clear that many women wanted to *"do the right thing by the baby"* (15), to ensure they had a healthy start in life. Women expressed fear about the potential to pass their ill health onto the baby. Most often, the initial response to their diagnosis was concern for the baby; "*oh my God…you know is the baby gonna be OK?"* (16). Some women expressed feelings of responsibility for their condition, and their desire to minimise the effect on their baby's health; *"I knew I had brought it on myself by being overweight…I felt very responsible"* (15).

Supportive environment: family

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When the familial support was positive, women were motivated to self-manage for reasons beyond the health of the baby: *"The biggest thing I can recommend is getting support. If you try to do it on your own, it's going to seem very confusing and tedious"* (17). When speaking about her partner one woman said *"he was very supportive in helping me get over things"* (18), due to her partner's support she was able to overcome her fear of blood glucose self-testing. Families and partners who adhered to the same self-management principles in their daily lifestyles helped women motivate themselves to follow dietary and exercise guidelines.

Supportive environment: health care professionals

Similarly, a supportive and constructive relationship with health care professionals proved to be a strong motivator for self-management, often minimising feelings of anxiety and uncertainty. Healthcare professionals who provided self-management advice gave women more autonomy over their pregnancy, *"the diabetes educator is really friendly…she did a demo in front of me, how to inject yourself"* (18). Women who understood their condition better were able to make positive choices about their self-management; *"I would limit my foods in order to control my blood sugar"* (19).

Barriers to self-management

Feeling anxious and overwhelmed

In this dataset, women often described feelings of anxiety and were overwhelmed by the prospect of managing their new condition. Women reported feelings of unhappiness or low mood while learning to self-manage, calling it "*frustrating*". Some women found trying to follow self-management guidelines created lot of anxiety: *"It creates a lot of anxiety. Like I am hungry but I don't want to eat now because I don't know if I harm the baby. But I feel hungry and could not eat"*

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(20). While it was often a lack of information that made women feel anxious, there was some evidence that being given *"too much information (about gestational diabetes)"* (15) also made women feel overwhelmed. Women were negative about self-management when they perceived that it had not gone to plan: *"When I took the higher dose [of insulin], I felt defeated…this is crazy, I don't think my body is ever going to be perfect"* (16).

Lack of knowledge and understanding

A lack of knowledge and understanding often led to a lack of motivation to self-manage. Women who did not understand their condition often underestimated the seriousness of their diagnosis or heard anecdotally from other women that it was not *"a severe illness"* (21) thus preventing women from committing to self-management. Contrastingly, others felt as though it was *"the end of the world"* (17) leading to women feeling too overwhelmed to self-manage efficiently. Central to the lack of understanding was a perception of insufficient education from health care professionals; one woman said that she was given *"a list of food but no instructions"* meaning *"you really have to work it out for yourself"* (15).

Lack of support: family

This dataset observed traditional forms of support networks: women's families, friends, and their healthcare professionals. A key theme to emerge from the data was the negative effect of a lack of familial support, which made women feel isolated and alone. Without strong family support networks women either stopped complying with self-management or lost the motivation to self-manage. Concerning diet self-management women reported their families being unwilling to follow the same dietary restrictions because *"you can't expect everybody to change everything"* (15), or failing to understand the importance of the dietary guidelines. Some women expressed that adhering to the strict diabetes diet made them feel socially isolated.

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Lack of support: health care professionals

A discordant relationship with healthcare professionals made women feel less motivated to self-manage, *"I love my OBGYN, but I feel they are always in a hurry. Like they don't have time to sit there and talk to you about what to do about it (GDM), but they are always in a hurry"* (17). One woman described her healthcare professional's attitude as *"here, you take the ball, and you run with it" (22),* but found that when her family were not informed about her self-management needs, it was difficult for her to stay on track. Women who wanted to self-manage reported either not being supported adequately by their healthcare professionals or subject to paternalistic responses: *"you have no idea what you're doing, I know best, here's what we're going to do"* (23). Some women had difficulty accessing information and understanding the self-management tasks, particularly when there were language barriers: *"It's a bit difficult because my English isn't very good…I have always been asking for an interpreter, but I have never got one"* (24). There were also concerns ranging from a lack of information (*"they didn't tell me what's the side effects for the baby"* (15)) to accusations of patronising healthcare professionals who *"talked down to [me]"* and *"didn't trust me"* (17).

Discussion

Principle Findings

This review found that women with diabetes predominantly undertake self-management during pregnancy for the health of their baby. Support networks are crucial in alleviating the burden that comes with managing a chronic condition, particularly a new one, such as gestational diabetes. When these networks are less effective, women may feel overwhelmed by the self-management, which is compounded by a lack of knowledge and understanding.

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One of the most notable findings from this review was the lack of research dedicated to managing chronic conditions in pregnancy. Diabetes was the only condition represented in the included literature, but with only 16 papers eligible for inclusion, the qualitative evidence base around managing gestational or chronic diabetes in pregnancy is still thin. While the findings of this review are framed by the self-management challenges of diabetes, a number of generic activities are applicable across other chronic conditions. Self-management of diabetes (type 1, type II and gestational) includes self-testing (of blood glucose levels) and diet and exercise management, all of which are applicable more widely in pregnancy. However, it is clear there is a lack of specific research on other chronic conditions in pregnancy, such as hypertension. It is also worth noting that in the general population chronic conditions are self-managed due to the absence of a 'cure', but this is not strictly the case in pregnancy where conditions can be temporarily bounded by the pregnancy; gestational diabetes and hypertension (or pre-eclampsia) can be resolved with delivery, and if they are not resolved, they are re-categorised as type 2 diabetes or hypertension (25) – thus, they are no longer conditions brought about by pregnancy.

While some of the motivations and barriers identified in this review are applicable to chronic conditions in the general population, this review has also demonstrated that the motivating factors for pregnant women to self-manage are different to those motivating the general population; none more so than the desire to improve outcomes for the baby. As far as the authors are aware this paper is the first to use the burden of treatment theory as a theoretical lens to explore experiences of chronic disease management in pregnancy. Although this theory does not fully account for the specific challenges of pregnancy, as demonstrated by this review, it still broadly accounts for the actors and work involved in self-management in this context. Using this theory allowed for a deeper understanding of the burden of self-management and the work involved.

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Interpretation

Acting for the Baby

May et al argue that the work of self-management in the general population is often done in the context of "*holding together something larger and more complex*" (14), alluding to everyday life. While this applies in pregnancy, the work of self-management is also done in the context of maintaining the health of the unborn baby. Previous literature has touched on pregnant people acting "*not for the self but also for the unborn "other"*' (26) and the burden of pregnant women having to be accountable for the health of their baby alongside their own health cannot be underestimated. Current self-management models are not equipped to consider that particular burden of treatment.

Support Networks

Current self-management models emphasise that "*self-management is manifested as both an individual and family construct*" (27) and underscore the importance and necessity of social networks. The importance of social networks is also realised in the Burden of Treatment theory. May asserts that "*knowledge and beliefs about health and healthcare are often shared [within social networks]…decisions about what to do, and how to access services, are often distributed amongst multiple participants in a social process*" (14). In pregnancy, the sharing of lay knowledge and expertise is commonplace. Hinton et al demonstrated the nature of information sharing among pregnant people, often now in the form of online communities, in those with pregnancy hypertension or raised blood pressure (28). Pregnant women with hypertension valued the advice and support of their peers, and in the online space peer advice tended to trump the advice of medical professionals (28). Support networks are a central part of any self-management model in the general population, often emphasising that "*self-management is manifested as both an*

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individual and family construct" (27). As both a barrier and a facilitator of self-management, it is evident family support is an important factor in continued adherence. While current self-management models do not accommodate pregnancy entirely, this review demonstrates that elements of existing models, such as social networks and the need for education around diagnosis, are applicable.

Strengths and Limitations

The research question evolved throughout the search as it became apparent that little data on self-management in pregnancy existed, and none outside of diabetes management. This review should act as a call for further work. Lived experiences are often over looked in self-management literature and by using qualitative literature we have been able to address some of these research gaps.

Limitations include that the coding framework is constrained by how the original researchers interpreted the interviews and their interactions with patients. Distinguishing between pre-existing chronic conditions and pregnancy acquired chronic conditions is important, but it was not a distinction that was reflected in the dataset; it is both a limitation of the included studies and this review. While the majority of the included papers were on gestational diabetes, there were two on pre-existing diabetes, and three with a mixed population of gestational and chronic diabetes. The burden of treatment theory does not perfectly map to pregnancy, however elements of this theory are helpful in understanding how and why patients interact with self-management. Many of the included papers specifically considered barriers to self-management, as opposed to facilitators, thus potentially creating a negative bias towards self-management.

Included studies took place across a variety of international settings, operating within different health systems, ranging from those dominated by private insurance-based systems to those with national healthcare systems aimed at universal healthcare coverage. Individualised medicine is

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intertwined with self-management; six of the included studies took place in the United States, where they operate within an insurance-based system that lends itself to an individualised approach to medicine. With the exception of Canada, New Zealand, and Singapore, none of the other countries included in this review have health systems based on almost near universal health care coverage. A patient's relationship with their healthcare system also affects their relationship with the acceptability of self-management and their efficacy undertaking it. As none of the included studies were conducted in the UK, it raises questions about how pregnancy self-management would operate within the NHS.

Conclusions

Clinical Implications

This review suggests there are several ways to improve adherence to self-management interventions in pregnancy, these interventions often involve self-monitoring and titrating medication. Health care professionals communicating the importance of self-management in terms that focus on the health of the baby, and educating women on the condition they have been diagnosed with, could have a positive impact on self-management adherence.

Research Implications

There is a need for research on the self-management of chronic conditions in pregnancy, particularly in chronic conditions other than diabetes. There are ongoing trials on self-management in pregnancy, predominantly in hypertension, but how self-management will fit into usual care remains to be seen. Remote forms of care are now more commonplace, in light of this, a better, qualitative understanding of the new burden of treatment self-management creates is needed.

Main Conclusions

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This review shows that the primary motivating factor for women self-managing is the health of their baby. Their support networks and their understanding of the condition contribute to whether they self-manage effectively. The burden of treatment shifting to women requires further research, as patient work increases, feelings of anxiety can also increase. In this dataset, it was clear that some women found this treatment shift to be overwhelming but their anxiety largely stemmed from a lack of knowledge and understanding of the condition with which they had been diagnosed. Evidently, education is a barrier to pregnant women effectively self-managing, as when these issues are addressed women are willing and able to self-manage.

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Contribution to Authorship

BJ, KT, LH, RM and NR contributed to creating the research question and parameters of the review, including the search strategy. BJ conducted the search with assistance from NR. BJ and JK reviewed the titles and abstracts of identified articles and carried out the full text assessments. Disagreements on reviewed articles were resolved by consensus or discussion with KT. Data extraction and complete analysis was conducted by BJ and reviewed by KT, LH and RM. The write up was led by BJ, with edits provided by KT, LH and RM.

Ethical approval

Not applicable.

Competing interests

The authors declare no conflict of interest.

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expressed in this publication are those of the authors and not necessarily those of the NHS, the NIHR, the Department of Health and Social Care, or the Primary Care Research Trust.

Patient Consent

Not required

Data Sharing Statement

All data relevant to the study are included in the article or uploaded as supplementary information.

Public and Patient Involvement

Patients and the public were not involved in this systematic review.

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Tables

Table 1: Characteristics of Included Studies

Title	Year	Authors	Chronic Condition	Journal	Sample Size	Setting	Qualitative Method
Living with gestational diabetes in a rural community	2014	Abraham , K. Wilk, N.	Diabetes	The American Journal of Maternal and Child Nursing	10	USA	One on one interviews
Women's experiences of gestational diabetes self-management: A qualitative study	2013	Carolan, M.	Diabetes	Midwifery	15	Australia Diabetes clinic	One on one interviews, focus group
Women's experiences of factors that facilitate or inhibit gestational diabetes self-management	2012	Carolan, M. Gill, G. Steele, C.	Diabetes	BMC Pregnancy and Childbirth	15	Australia Diabetes clinic	One on one interviews, focus group
Experiences, perceptions and self-management of gestational diabetes in a group of overweight multiparous women	2014	Chavez-Courtois, M. Graham, C. Romeo-Perez, I. Sanchez-Miranda, G. Sanchez-Jimenez, B. Perichart-Perera, O.	Diabetes	Cien Saude Colet	5	Spain	One on one interviews
'Is gestational diabetes a severe illness?' exploring beliefs and self-care among women with gestational diabetes living in a rural area of the south east of China	2016	Ge, L. Wikby, K. Rask, M.	Diabetes	Australian Journal of Rural Health	17	China Obstetric clinics, wards	One on one interviews
The maternal experience of having diabetes in pregnancy	2011	Nolan, J. McCrone, S. Ilana, R. Chertok, A.	Diabetes	Journal of the American Academy of Nurse Practitioners	8	USA	Focus groups
Self-monitoring lifestyle behaviour in overweight and obese pregnant women: qualitative findings	2018	Sheih, C. Draucker, C.B	Overweight/obese	Journal of Clinical Nursing	13	USA Clinics (community health centre and health network)	One on one interviews
Socio-cultural aspects of self-management in gestational diabetes	2012	Wazqar, D. Evans, M.	Diabetes	Diabetes Nursing	12	Canada	Secondary analysis of one on one interviews
'I was tired of all the sticking and poking': Identifying barriers to diabetes self-care among low-income pregnant women	2015	Yee, L. McGuire, J. Taylor, S. Niznik, C. Simon, A.	Diabetes	Journal of Healthcare for the Poor and Underserved	10	USA Clinic	One on one interviews
Factors promoting diabetes self-care among low-income, minority pregnant women	2016	Yee, L. McGuire, J. Taylor, S. Niznik, C. Simon, A.	Diabetes	Journal of Perinatology	10	USA Diabetes clinic	One on one interviews
Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: a phenomenological research	2017	Youngwanichsetha, S. Phumdoung, S.	Diabetes	Journal of Clinical Nursing	30	Thailand Antenatal clinic, diabetes clinic, obstetric ward	One on one interviews
Self-management of gestational diabetes among Chinese migrants: A qualitative study	2018	Wah, Y. McGill, M. Wong, J. Ross, G. Harding, AJ. Krassa, I.	Diabetes	Women and Birth	18	Australia Antenatal clinic	One on one interviews

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Ethnic Differences in Dietary Management of Gestational Diabetes Mellitus: A Mixed Methods Study Comparing Ethnic Chinese Immigrants and Australian Women	2019	Wan, C. Teede, H. Nankervis, A. Aroni, R.	Diabetes	Journal of the Academy of Nutrition and Dietetics	83	Australia Maternity services, tertiary hospitals	One on one interviews and diary entries
The experience of gestational diabetes for indigenous Māori women living in rural New Zealand: qualitative research informing the development of decolonising interventions	2018	Reid, J. Anderson, A. Cormack, D. Reid, P. Harwood, M.	Diabetes	BMC Pregnancy and Childbirth	10	New Zealand Primary care clinic	One on one interviews
"Diabetes Just Tends to Take Over Everything": Experiences of Support and Barriers to Diabetes Management for Pregnancy in Women With Type 1 Diabetes	2019	Singh, H. Ingersoll, K. Gonder-Frederick, L. Ritterband, L.	Diabetes	Diabetes Spectrum	15	USA Diabetes clinic	One on one interviews
Barriers to Gestational Diabetes Management and Preferred Interventions for Women With Gestational Diabetes in Singapore: Mixed Methods Study	2020	Hewage, S. Audimulam, J. Sullivan, E. Chi, C. Yew, T. W. Yoong, J.	Diabetes	JMIR Formative Research	15	Singapore Gestational Diabetes clinic	One on one interviews, in a mixed methods study also including a survey

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Table 2: Quote Table: Barriers and Motivations for Self-management

Barriers to self-management		Motivations for self-management	
Lack of knowledge and understanding	"Actually, I didn't know anything about it...So, I...thought that if you were diabetic then you kind of get it. But then I didn't know something you can just develop during pregnancy as well. So, it was quite new to me." (29)	**Desire for healthy baby**	"I have a responsibility to care for my child…and to care for this baby inside of me…For me, I wasn't going to let anything get in the way…" (30)
Feeling anxious and overwhelmed	"With the fourth pregnancy, I had sort of lost control. I was living in the garage with my three children [and husband]. I was huge and six-months [along in my pregnancy] when I was diagnosed with depression….I didn't want to do anything. I'd just wake up [and] went through the motions." (31)	**Supportive environment: family**	"If it's not because of my husband, I couldn't have made it this far, like he would remind me, sometimes when I want to eat something he would just remind me not to, and like when we go for shopping he would buy the healthy stuff." (24)
Lack of support: family	"I need to cook for my husband and I can't make him eat the same thing…I need to follow him because he needs energy for his work…and I can't avoid using sauce, you know how on the menu it says you need to avoid sauces and stuff, I can't do that, my husband wouldn't want to eat according to that." (24)	**Supportive environment: HCP**	"The diabetes educator is really friendly…she explained things very like, in a very good way Yes, yes and – like, she did a demo in front of me, how to inject yourself. It was really scary first time. They told me everything." (19) "The first pregnancy I had a lady (educator) that was really excellent, very understanding, very approachable. She would listen to my concerns and we had a conversation rather than just a one-way flow of information. And so I had a very positive experience with her." (15)
Lack of support: HCP	"You are told that you have this and nobody spends time with you, it's kind of frustrating because you want to cry. You think that it's something you did wrong" (17) "According to my experience, what they (health professionals) discussed are all about the dietary habit of women here (in Australia), so if the suggestions could be more relevant to Chinese lifestyle, a lot of things that have been taught currently are not meaningful." [20]		

Figures

Figure 1: PRISMA Flow Diagram

Legend: PRISMA Flow Diagram of study search strategy

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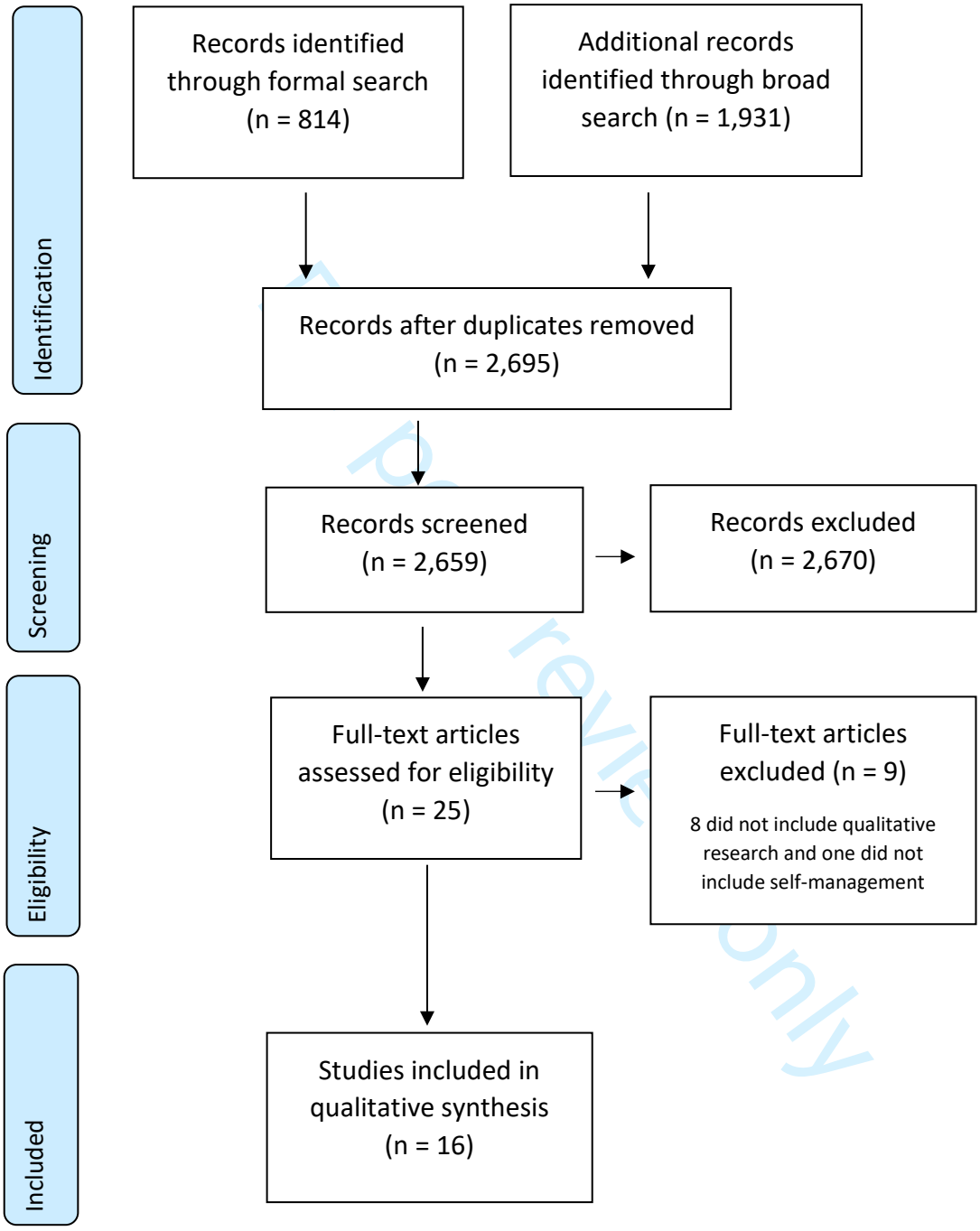
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Identification
Screening
Eligibility
Included
Records identified through formal search (n = 814)
Additional records identified through broad search (n = 1,931)
Records after duplicates removed (n = 2,695)
Records screened (n = 2,659)
Records excluded (n = 2,670)
Full-text articles assessed for eligibility (n = 25)
Full-text articles excluded (n = 9)
8 did not include qualitative research and one did not include self-management
Studies included in qualitative synthesis (n = 16)

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Systematic review

1. * Review title.

Give the title of the review in English

What are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?

2. Original language title.

For reviews in languages other than English, give the title in the original language. This will be displayed with the English language title.

3. * Anticipated or actual start date.

Give the date the systematic review started or is expected to start.

01/04/2019

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31/12/2019

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Formal screening of search results against eligibility criteria	Yes	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Provide any other relevant information about the stage of the review here.

6. * Named contact.

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Beth Jakubowski

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

Ms Jakubowski

7. * Named contact email.

Give the electronic email address of the named contact.

bethany.jakubowski@phc.ox.ac.uk

8. Named contact address

Give the full institutional/organisational postal address for the named contact.

9. Named contact phone number.

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Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. **NOTE: email and country now MUST be entered for each person, unless you are amending a published record.**

Ms Bethany Jakubowski. University of Oxford
Ms Jaspreet Khaira. University of Oxford
Professor Richard McManus. University of Oxford
Dr Katherine Tucker. University of Oxford
Dr Lisa Hinton. University of Cambridge

12. * Funding sources/sponsors.

Details of the individuals, organizations, groups, companies or other legal entities who have funded or sponsored the review.

CLAHRC Oxford

Grant number(s)

State the funder, grant or award number and the date of award

13. * Conflicts of interest.

List actual or perceived conflicts of interest (financial or academic).

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country must be completed for each person, unless you are amending a published record.**

15. * Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

What are the current behaviours and attitudes among pregnant women regarding the self-management and self-monitoring of chronic conditions in pregnancy?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

The search terms used to find articles in PubMed

- Pregnancy MeSH
- Pregnant women MeSH
- Pregnan* OR gestational OR maternal
- Decision making MeSH
- Self-care MeSH
- Shared decision making OR self manag* OR self-monitor

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The search terms used for CINAHL

- Pregnancy OR Expectant Mothers

- Pregnan* OR gestational OR maternal

- Self-care

- Shared decision making OR self-manag* OR self-monitor

There was also a broad secondary search using the search term 'pregnancy self-management' in PubMed

17. URL to search strategy.

Upload a file with your search strategy, or an example of a search strategy for a specific database, (including the keywords) in pdf or word format. In doing so you are consenting to the file being made publicly accessible. Or provide a URL or link to the strategy. Do NOT provide links to your search **results**.

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Do not make this file publicly available until the review is complete

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

This review is on the self-management of chronic conditions in pregnancy i.e. hypertension, diabetes, kidney disease. The scope of the review is to look at the feasibility and acceptability of women self-managing, and how they regard this new burden of treatment.

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

The population being studied is pregnant women with a chronic condition, there are no further restrictions on the study population.

Inclusion Criteria

1. Chronic condition: diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease, others to be reviewed as they come up

2. Pregnancy

3. Self-management: self-monitoring, self-weighing, self-care, others to be reviewed as they come up

4. Qualitative.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The

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preferred format includes details of both inclusion and exclusion criteria.

This review is looking at the self-management of chronic conditions in pregnancy. Self-management can be defined as self-monitoring (i.e. of blood glucose or blood pressure), or as action on the basis of self-monitoring (i.e. diet management, self-care, weight management).

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

Usual care if appropriate.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Qualitative studies are eligible for inclusion, including secondary analysis of previous qualitative research.

Inclusion Criteria

1. Chronic condition: diabetes (gestational or chronic), hypertension (gestational or chronic), obesity, kidney disease, others to be reviewed as they come up

2. Pregnancy

3. Self-management: self-monitoring, self-weighing, self-care

4. Qualitative

23. Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

Chronic conditions in pregnancy are under researched, so the aim of this review is to examine the current qualitative evidence of the behaviours and attitudes towards the self-management of chronic conditions in pregnancy. It is important to determine what, if any, self-management currently takes place and whether women find it acceptable and feasible to self-manage. Therefore the inclusion criteria for this review is for the participant to have a chronic condition (i.e. hypertension, diabetes) during pregnancy (this can be defined as gestational or chronic), and for some form of self-management to be taking place. This self-management can either be self-monitoring (i.e. of blood glucose or blood pressure), or action on the basis of self-monitoring (i.e. diet management, self-care, weight management).

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

The study aims to explore whether self-management of chronic conditions in pregnancy is acceptable and

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feasible to pregnant women. It also aims to explore current practice and the current evidence base of chronic disease management in pregnancy.

Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

None.

Measures of effect

Please specify the effect measure(s) for you additional outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

All of the studies identified in the search were screened independently by two reviewers (title and abstract) who then came together to agree on which articles should get a full text review. Both reviewers then independently completed the full text review and met again to agree on which articles should be included in the final review. Discrepancies were resolved by reviewing the definition of self-management and chronic condition as these were often the causes of uncertainty. The first author will then use a data extraction spreadsheet to list the key themes and findings from the papers. 10% of the included articles will be screened by another member of the study team, measuring the 10% of articles against the inclusion criteria and the COREQ quality assessment guidelines. The included studies will then be put into NVivo 12 and a thematic analysis, using the guidelines produced by Braun and Clarke (2006), will be carried out by the first author. A theoretical lens will be used to reflect the themes that emerge during data analysis.

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

Two researchers carried out the screening of articles, and met after the title and abstract screening to feed back results and decide which articles required a full text review. They met again following this screening to decide which articles should be included in the final review. To determine the quality of the studies, the COREQ guidelines will be used, which is a standard set of guidelines to determine the quality of qualitative work. The quality of the studies will be acknowledged in the write up, in order to be transparent about the

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quality of the work in this field to date.

28. * Strategy for data synthesis.

Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta-analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.

Using NVivo 12, an initial thematic analysis will be conducted by the first author to create a coding framework of themes and sub themes. A theoretical lens will be used to develop the coding framework further. Discrepancies will be discussed with the wider study team and feedback will be incorporated into the framework.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

Not currently planned

30. * Type and method of review.

Select the type of review, review method and health area from the lists below.

Type of review

Cost effectiveness
No

Diagnostic
No

Epidemiologic
No

Individual patient data (IPD) meta-analysis
No

Intervention
No

Meta-analysis
No

Methodology
No

Narrative synthesis
No

Network meta-analysis
No

Pre-clinical
No

Prevention
No

Prognostic

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No

Prospective meta-analysis (PMA)
No

Review of reviews
No

Service delivery
No

Synthesis of qualitative studies
Yes

Systematic review
Yes

Other
No

Health area of the review

Alcohol/substance misuse/abuse
No

Blood and immune system
No

Cancer
No

Cardiovascular
No

Care of the elderly
No

Child health
No

Complementary therapies
No

COVID-19
No

Crime and justice
No

Dental
No

Digestive system
No

Ear, nose and throat
No

Education
No

Endocrine and metabolic disorders
No

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NHS
National Institute for
Health Research

Eye disorders
No

General interest
No

Genetics
No

Health inequalities/health equity
No

Infections and infestations
No

International development
No

Mental health and behavioural conditions
No

Musculoskeletal
No

Neurological
No

Nursing
No

Obstetrics and gynaecology
No

Oral health
No

Palliative care
No

Perioperative care
No

Physiotherapy
No

Pregnancy and childbirth
Yes

Public health (including social determinants of health)
No

Rehabilitation
No

Respiratory disorders
No

Service delivery
No

Skin disorders
No

Social care

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No

Surgery
No

Tropical Medicine
No

Urological
No

Wounds, injuries and accidents
No

Violence and abuse
No

31. Language.

Select each language individually to add it to the list below, use the bin icon to remove any added in error.
English

There is not an English language summary

32. * Country.

Select the country in which the review is being carried out. For multi-national collaborations select all the countries involved.

England

33. Other registration details.

Name any other organisation where the systematic review title or protocol is registered (e.g. Campbell, or The Joanna Briggs Institute) together with any unique identification number assigned by them. If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34. Reference and/or URL for published protocol.

If the protocol for this review is published provide details (authors, title and journal details, preferably in Vancouver format)

Add web link to the published protocol.

Or, upload your published protocol here in pdf format. Note that the upload will be publicly accessible.

No I do not make this file publicly available until the review is complete

Please note that the information required in the PROSPERO registration form must be completed in full even if access to a protocol is given.

35. Dissemination plans.

Do you intend to publish the review on completion?

Yes

Give brief details of plans for communicating review findings.?

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We will aim to publish the review in a high impact scientific journal.

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords help PROSPERO users find your review (keywords do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

Pregnancy; maternal health; chronic conditions; self-management

37. Details of any existing review of the same topic by the same authors.

If you are registering an update of an existing review give details of the earlier versions and include a full bibliographic reference, if available.

38. * Current review status.

Update review status when the review is completed and when it is published.New registrations must be ongoing so this field is not editable for initial submission.
Please provide anticipated publication date

Review_Ongoing

39. Any additional information.

Provide any other information relevant to the registration of this review.

40. Details of final report/publication(s) or preprints if available.

Leave empty until publication details are available OR you have a link to a preprint (NOTE: this field is not editable for initial submission). List authors, title and journal details preferably in Vancouver format.

Give the link to the published review or preprint.

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Appendix 1: Search Strategy

Two databases were searched, CINAHL and Pub Med. The search terms used to find articles in Pub Med (n=107) were MeSH terms 'pregnancy,' 'pregnant women', 'decision making', 'self-care', and 'pregnan* OR gestational OR maternal', and 'shared decision making OR self manag* OR self-monitor'. There was also a broad secondary search in Pub Med using the search term 'pregnancy self-management' which identified a further 1,931 articles for review. The search terms used for CINAHL (n=707) were 'pregnancy OR expectant mothers', 'pregnan* OR gestational OR maternal', 'self-care', and 'shared decision making OR self-manag* OR self-monitor'. This identified a total of 2,745 articles, which when controlled for duplicates left 2,695 for review. 25 articles were identified for full text review, of which 16 were selected for the systematic review.

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Quality Appraisal Checklist	Personal Characteristics				Relationship with Participants			Theoretical Framework	Participant Selection				Setting			Data collection						
Title	Interviewer/Facilitator	Credentials	Occupation	Experience and training	Relationship established	Participant knowledge of the interviewer/interview	Interviewer characteristics	Methodological orientation and theory	Sampling	Method of approach	Sample size	Non-participation	Setting of data selection	Presence of non-participants	Description of sample	Interview guide	Repeat interviews	Audio/visual recording	Field notes	Duration	Data saturation	Transcripts returned
Self-monitoring lifestyle behaviour in overweight and obese pregnant women: qualitative findings	First author	Author 1: DNSc, MPH, RNC-OB, FAAN. Author 2: PhD, RN, FAAN	Author 1: Associate Professor. Author 2: Researcher	First author: Expert in prenatal health promotion and content analysis. Second author: PhD. Doctoral student carrying out analysis: undergraduate degree in exercise science, experience in a home visitation programme for low-income pregnant women	First author called or e-mailed each woman 3 - 7 days before a scheduled interview	Not mentioned - assumed through recruitment	No bias reported	Self-monitoring theory, behavioural and lifestyle change theories	Purposive	Interview explained to trial participants and informed consent sought	13	22 women enrolled in the intervention, 9 refused interview	10 interviewed in clinic private rooms, 3 over phone	No	Mean age 29.5 years, majority obese, majority were African American (77%), unmarried (69%), previous pregnancy (77%), third trimester (69%)	Yes	No	Audio	No	30 minutes	Not discussed	No
"I was tired of all the sticking and poking": Identifying Barriers to Diabetes Self-Care Among Low-Income Pregnant Women	Author 1 and 2	Author 1: MD, MPH, Author 2: MD, MPH, Author 3: MPH, Author 4: APN, CDE, Author 5: MD, MPH	Author 1: Fellow in Maternal-Fetal Medicine. Author 2: Resident in OBGYN at Northwestern. Author 3: Senior Research Coordinator at Northwestern. Author 4: Advance Practice Nurse in Maternal-Fetal Medicine. Author 5: Professor of OBGYN at Northwestern	Extensive expertise in maternal-fetal medicine and research	Not mentioned	Were identified because they had GDM so knew why the research was being conducted	No bias reported	Cognitive load theory	Purposive	At a clinic serving public-assisted low-income women. Informed consent sought.	10	46 women identified, 22 ineligible, 12 declined participation	Conducted in private	No	Mean age 28.5, African American (4), Latina (4), Native American (1), Asian (1). Most educated to high school degree level, most multiparous, all diabetic	Yes	Three interviews per person	Audio	No	30-60 minutes	Reached at 29 interviews	No
The maternal experience of having diabetes in pregnancy	Not reported by coder/analyser was present for all interviews and focus groups	Author 1: MD, Author 2: PhD, RN, PMHCNS-BC, Author 3: PhD, RN, IBCLC	Not reported	Extensive expertise judging by credentials	Not mentioned	Recruitment flyers provided some knowledge of subject matter	No bias reported	Phenomenology	Convenience	Flyers distributed in healthcare provider offices	8	Not reported	Over the phone or in person	Yes, additional investigator to take field notes	Average age 35, mostly highly educated, no demographics reported	Yes	No - these were focus groups	Audio	Yes	Not reported	Not discussed	Some were to check participants views were represented correctly
Women's experiences of gestational diabetes self-management: a qualitative study	Not reported	Author 1: PhD	Associate Professor Midwifery	Not reported but has a PhD in a related field	Approached by researcher prior to interview to confirm interest	Recruited from diabetes clinic fitting criteria	No bias reported	Phenomenology	Purposive (blinded for review)	Women invited to participate if fitted criteria, approached by researcher if Diabetes Coordinator confirmed their interest	15	Not reported	10 phone interviews, one in interviewee's home, one focus group at the hospital clinic	No	Variety of ethnic backgrounds, 23 - 40, majority high school educated, majority primiparous	Not reported	No	Audio	No	Not reported	Not discussed	No
Factors promoting diabetes self-care among low-income, minority pregnant women	Author 1 and 2	Author 1: MD, MPH, Author 2: MD, MPH, Author 3: MPH, Author 4: APN, CDE, Author 5: MD, MPH	Author 1: Fellow in Maternal-Fetal Medicine. Author 2: Resident in OBGYN at Northwestern. Author 3: Senior Research Coordinator at Northwestern. Author 4: Advance Practice Nurse in Maternal-Fetal Medicine. Author 5: Professor of OBGYN at Northwestern	Members of the research team reported to have prior qualitative research experience and extensive interviewing training	Not mentioned	Were identified because they had GDM and at their first clinic visit so knew why the research was being conducted	No bias reported	Cognitive load theory	Purposive	Recruited at their first clinic visit for diabetes care	10	46 women identified, 22 ineligible, 12 declined participation	Conducted in private	No	Mean age 28.5, African American (4), Latina (4), Native American (1), Asian (1). Most educated to high school degree level, most multiparous, all diabetic	Yes, provided in article	Three interviews per person	Audio	No	30-60 minutes	Reached at 29 interviews	No
Socio-cultural aspects of self-management in gestational diabetes	Author 1	Author 1: PhD student. Author 2: Assistant Professor	Author 1: PhD student. Author 2: Assistant Professor Nursing	PhD student	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Phenomenology	Purposive	Women being treated for GD were recruited	12	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Aged between 23 - 38, most worked outside the home, most were married, mix of first pregnancy and second pregnancies	Yes	Two interviews conducted per person	Audio	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis	Not mentioned - secondary analysis
Women's experiences of factors that facilitate or inhibit gestational diabetes self-management	Author 1	Not reported	Not reported	Not reported	Access to women facilitated by diabetes co-ordinator, no relationship pre-established with researcher	Recruited through diabetes clinic and by diabetes co-ordinator so would have been told what the study was about	No bias reported	Interpretative phenomenology	Purposive	Diabetes coordinator approached women, participants were recruited after they attended GDM education and had at least 3 weeks self management of condition	15	30 women met inclusion criteria, 20 agreed to participate, 15 available on day of interview	Focus group in private room in clinic, phone interviews, face to face interview at one woman's home	No	Age range 23 -40, education level lower than population average, mix of ethnic backgrounds, majority of women primiparous	Not reported	No	Audio	No	Not reported	Not mentioned	No
Lived experience of blood glucose self-monitoring among pregnant women with gestational diabetes mellitus: a phenomenological research	Not reported	Author 1: PhD, RN, Assistant Professor of Nursing. Author 2: PhD, RN, Associate Professor of Nursing	Author 1: Assistant Professor. Author 2: Associate Professor	Not reported	Prolonged engagement was carried out by researchers to build a rapport with participants before interviews	Participants approached by staff nurses so knew about what the study was about prior to taking part	No bias reported	Phenomenology	Purposive	Approached at antenatal care clinics and informed consent obtained	30	Not reported	Interviewed at the antenatal clinic, diabetes clinic or obstetric ward	No	The mean age of participants 32.5 years, 17 women were Buddhists, 13 women were Muslims. Majority multiparous and employed.	Yes	Two interviews conducted per person	Audio	No	30 - 45 minutes	Data were collected until saturation (after six months)	No
Living with gestational diabetes in a rural community	Not reported	Author 1: DNP, APRN-C. Author 2: DNS, WHNP-C	Author 1: Visiting Assistant Professor of Nursing. Author 2: Visiting Assistant Professor	Not reported	Established through recruitment	Had diagnosis of the disease which was being explored in interviews, and recruited via study posters	No bias reported	Phenomenology	Purposive and snowball	Information about study posted in obstetric and health care provider officers in New York State	10	Not reported	Interviewed in person or at phone	No	All women Caucasian between 25-49. All had history of GDM.	Yes	Follow up interviews conducted with four participants	Audio	No	15 - 30 minutes, second interviews 15 - 30 minutes	Checked by sharing core themes with participants	Transcripts of first interview provided to participants doing a second interview for verification
"Is gestational diabetes a severe illness?" Exploring beliefs and self-care behaviour among women with gestational diabetes living in a rural area of the south east of China	Author 1	Author 1: M, Msc. Author 2: RN, PhD. Author 3: RN, PhD	Author 1 a teacher of maternal care and bilingual	Not reported	Not mentioned	Interviewer did not know participants prior to study, was not part of the clinical care team.	No bias reported	No specific theoretical framework reported	Purposive	Women purposefully recruited from obstetric clinics and wards at a specific hospital.	17	Not reported	Interviewed in a room at the hospital	No	27.5 median age, six women high educational level, five middle educational level, six low educational level	Yes and provided in paper	No	Audio	No	40 - 60 minutes	Not mentioned	No
Experiences, perceptions and self management of gestational diabetes in a group of overweight multiparous women	Not reported	Not reported	Not reported	Interviews conducted by experts in anthropology and social psychology	Established through recruitment and screening process	Women were given a health questionnaire prior to recruitment	No bias reported	Phenomenology	Purposive	Women recruited through department of diabetes, given questionnaire and selected based on the questionnaire results	5	Not reported	Not reported	No	Reported in table	Yes	No	Audio	Yes	90 minutes	Not mentioned	No
Ethnic Differences in Dietary Management of Gestational Diabetes Mellitus: A Mixed Methods Study Comparing Ethnic Chinese Immigrants and Australian Women	Not reported	Author 1: MN, APD Author 2: PhD Author 3: MD Author 4: PhD	Not reported	Not reported	No	Were identified because they had GDM so knew why the research was being conducted	No bias reported	Symbolic interactionist framework and grounded theory	Purposive	Women recruited through GDM clinics	83	11 declined to participate and reasons were given	Not reported but some on the phone (others in person)	No	Reported in table	Yes	No	Audio	Yes	45 - 90 minutes	Not mentioned but large sample size so assumed there was saturation	No
Self-management of gestational diabetes among Chinese migrants: A qualitative study	Chief Investigator	Not reported	Faculties reported	CI was fluent in English, Mandarin and Cantonese	Participant's had interviewer's contact details	Interested participants were provided with written information that explained the background of the study and provided the contact details of the researchers. All participants were recruited after they had received diabetes education from a diabetes educator and had at least two weeks of self-management experience.	No bias reported	Thematic analysis	Purposive	Between August and October 2016, participants were purposively recruited from the antenatal clinic at the RPAH in the Inner West Region of Sydney, Australia. Bilingual flyers about the study were distributed at the clinic. Interested participants were provided with written information that explained the background of the study and provided the contact details of the researchers.	18	Not reported	The interviews were conducted by the chief investigator in either a staff meeting room or a waiting room at the hospital, according to the participant's preference.	Yes, participant's partner present in two interviews	Reported in table: The mean age of participants was 34 years (SD = 3.6; range 29–41). The majority of participants came from China, Hong Kong or Taiwan and held a university degree	Yes	No	Audio	No	Not reported	Not reported	No
The experience of gestational diabetes for indigenous Māori women living in rural New Zealand: qualitative research informing the development of decolonising interventions	Not reported	Not reported	Faculties reported	Not reported	Lead researcher spoke to participants ahead of the interviewer	The contact details of all consenting participants were then provided to the lead researcher, who contacted participants via phone or home visits (when no phone service was available). The lead investigator provided detailed information about the research, answered any questions from women and obtained written informed consent and confidentiality agreements (including accessing their diabetes-related health data from the clinic) for each participant.	No bias reported	Kaupapa Maoris Methodology	Purposive - a sub sample of a larger study	The ten participants advised that they either had a clinical history of GDM (n = 8) or had been exposed to diabetes in utero (n= 2). Eligible participants were recruited via the local primary care clinic between September and December of 2015.	10	Reported in the larger study (6 withdrew)	Interviews undertaken in participants' homes or the health centre, or a mutually agreed venue.	No	10 Māori women aged between 30 and 69 years. Outlined in full table, participants had diverse pregnancy experiences and current diabetes-related health conditions	Yes	No	Audio	No	Not reported	Not discussed	Participants were invited to review, edit or change their transcript prior to data analysis.
"Diabetes Just Tends to Take Over Everything": Experiences of Support and Barriers to Diabetes Management for Pregnancy in Women With Type 1 Diabetes	Not reported	Not reported	Not reported	The main interviewer had a background in health psychology and skills in motivational interviewing.	Potential participants could indicate interest by contacting the investigator. Study procedures and the consent process reviewed with them.	Potential participants were informed that the purpose of the study was to identify barriers to and supports for blood glucose management around pregnancy	No bias reported	Thematic analysis	Purposive	Participants were recruited using advertisement flyers that were sent for distribution to diabetes specific clinics around the United States. Adult U.S. women (≥18 years of age) with type 1 diabetes who were fluent in English and were either pregnant, planning pregnancy for the immediate future, or had experienced a previous pregnancy were invited to participate in a phone interview.	15	27 indicated interest	Decided to conduct the interviews over the phone unless participants were local to the study site and wished to come to the study office	No	All participants were married, with a mean age of 34 years and a mean duration of type 1 diabetes of 21 years. Ten participants used a continuous glucose monitoring (CGM) system, and 13 used an insulin pump to manage their diabetes. Ten women had previously been pregnant, three were pregnant at the time of the interview, and five were trying for or contemplating pregnancy in the immediate future (within 0–4 months from the time of their interview).	Yes	No	Audio	Yes - notes and reflections recorded by interviewer	Between 30 and 50 minutes	To achieve data saturation, 10–15 people were recruited, with the possibility of conducting additional interviews, if necessary	No
Barriers to Gestational Diabetes Management and Preferred Interventions for Women With Gestational Diabetes in Singapore: Mixed Methods Study	First and second author	Author 1, PhD, Author 2 MPH, Author 3 MPH, Author 4 MBBS, MD, Author 5, MBBS, Author 6, PhD	Faculties reported	Not reported	Established through recruitment	Recruited from diabetes clinic fitting criteria, participants had previously participated in a survey on the same subject.	No bias reported	Thematic analysis	Purposive	Potential participants for the survey (this is a mixed methods study and interview participants were recruited from survey participants) were approached in the waiting room of the GDM clinic. The study team refrained from approaching women who seemed tired or anxious. After getting informed consent, women were given a self-administered survey. For pregnant women who were willing to participate in prenatal interviews, sessions were scheduled based on their convenience. In the second phase, to increase survey completion, the women were given an incentive of $25 (US $17.5). In the first phase, 50 patients participated in the survey.	15	Of the 50 patients who completed the survey, 35 gave verbal consent to participate in the interviews and 15 participated in the interviews	Most interviews were conducted at the health care facility or at the participants' homes, in a private space conducive for the participant to share thoughts effectively.	No	The women who participated in interviews had a demographic profile similar to that of the survey questionnaire participants. Half of the 15 participants in the age group of 31 to 35 years were primiparous or had a family history of T2DM, and most of them were Chinese or Indian, employed full time, with a degree, and attending a private prenatal clinic.	Yes	No	Audio	No	Not reported	Reported at 15 interviews	No

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Data analysis					Reporting					Reasons
Number of data coders	Description of coding tree	Derivation of themes	Software	Participant checking	Quotations presented	Data and findings consistent	Clarity of major themes	Clarify of minor themes	Quality Assessment	
2	Yes	Themes derived from data	N/A	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	Medium	Small sample size, short interviews, however themes were clear and recruitment process was transparent
2	Yes	Themes derived from data	ATLAS.ti 6	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	High	Despite small sample size, multiple interviews allowed for a richer dataset.
1 but checked informally by members of research team	No	Themes derived from data and then refined based on existing evidence of topic	N/A	Yes	Some reported, no quote table	Yes	Clear	Clear	Medium	Small sample size, even though focus groups tend to be better with a smaller sample size, this was still quite low and does not allow for a rich data set and exploration of themes
1	No	Themes derived from data	N/A	No	Yes with a quote table	Yes	Clear	Clear	High	Good sample size, clear discussion of themes and recruitment
2	Yes	Themes derived from data	ATLAS.ti 6	No	Yes in full quote tables by theme and sub theme	Yes	Clear	Clear	High	Despite small sample size, multiple interviews allowed for a richer dataset.
1	Yes	Themes derived from data	Not mentioned	Not mentioned - secondary analysis	No quote table, in text quotes included	Yes	Clear	Clear	Medium	A good sample size for a secondary analysis, but more information on primary data collection needed for transparency
1	Yes	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	High	Good sample size and mixed qualitative methods, leading to an apparently rich dataset. Paper was clear.
1	Yes	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	High	Good sample size , rich dataset and exploration of themes. Clear paper.
Not reported	No	Themes derived from data	Not mentioned	Yes participants were informed of core themes for member checking and saturation verification	No quote table, in text quotes included	Yes	Clear	Clear	Medium	Small sample size, but the discussion of themes was nuanced and rich. The interviews were also reported as being quite short.
Not reported	No	Themes derived from data	Not mentioned	No	No quote table, in text quotes included	Yes	Clear	Clear	Medium	Good sample size with rich, longer form interviews. Clear paper and recruitment process clearly explained.
Not reported	No	Themes derived from data	ATLAS.ti V.5	No	No quote table, in text quotes included	Yes	Clear	Clear	Low	Very small sample size, potentially too low to make any assumptions about experiences of self-management
At least 2	No	Themes derived from data	Nvivo 11	No	Yes and quote tables included	Yes	Clear	Clear	High	Very large sample size, very rich dataset, clear paper and all processes communicated well
2	No	Themes derived from data	Not mentioned	No	Yes, embedded in article	Yes	Clear	Clear	High	Good sample size, clear paper and clear recruitment and analysis processes described
The lead researcher independently coded the data, and discussed the codes and themes with two experiences kaupapa Māori researchers, who had access to the interview data	No	Themes derived from data	Not mentioned	Participants were invited to review, edit or change their transcript prior to data analysis.	Yes, embedded in article	Yes	Clear	Clear	Medium	Small sample size, but an exploratory study and acknowledged it was a sub set of women who are rarely interviewed about their experiences.
Each transcript reviewed and coded independently by two researchers	Yes	Themes derived from data	Not mentioned	No	Yes, embedded in article	Yes	Clear	Clear	High	Good sample size, clear description of analysis process and themes very clear and supplemented with quotes
1	Yes	Themes derived from data	Atlas Ti	No - but transcripts checked by an author for accuracy	Yes but quote tables not included	Yes	Clear	Clear	Medium	Based on the qualitative interviews alone the sample size is small. However, there was a rich discussion of themes and analysis was rigorous.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3, 4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5, 6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5, 6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7, 8, 10, 25, 26
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7, 8, 10, 25, 26
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7, 8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	N/A
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9, 10, 11

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10 - 14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15, 16, 17
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	17, 18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	18, 19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	19, 20

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